

Invisible Imports: The De Minimis Loophole and Tariff Evasion in the U.S.-China Trade War*

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Abstract

Firms responded to the U.S.–China trade war by exploiting tariff avoidance channels, including small-package *de minimis* shipments and rerouting through third countries. Under U.S. law, shipments valued below \$800 are exempt from tariffs and standard customs procedures and are not fully captured in official trade statistics.¹ Using shipment-level data from U.S. Customs and Border Protection, this paper constructs a new classification of products exposed to *de minimis* trade prior to the trade war. Combining this measure with a monthly HS10 country–product panel from 2017 to 2019, the analysis examines how tariff exposure reshaped trade flows across shipment and customs regimes.

U.S. imports from China declined sharply following the introduction of Section 301 tariffs. A 25 percentage point tariff increase implies roughly a 33% decline in import values for non-*de minimis* goods and a 39% decline for *de minimis*-exposed products. Measured tariff incidence and pass-through were significantly weaker for *de minimis* products, while rerouting through third countries increased primarily for non-*de minimis* goods. Together, the results suggest that firms adjusted along multiple margins, creating gaps between observed trade flows, tariff collection, and underlying import activity.

A structural framework of export channel choice formalizes firm behavior across direct shipment, *de minimis* entry, rerouting, and exit. The findings show how fragmented parcel trade and modern logistics systems can weaken the effectiveness, enforcement, and measurement of trade policy.

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¹In May 2025, the United States eliminated *de minimis* eligibility for imports from China and Hong Kong before extending the policy to imports from the rest of the world in August 2025.

1 Introduction

The U.S.–China trade war introduced sharp distortions into global supply chains, prompting a wide range of firm responses that circumvented the direct incidence of tariffs. Starting in 2018, the United States imposed new tariffs on approximately \$250 billion of Chinese imports, expanding by 2019 to cover over \$360 billion in goods. These duties reached rates as high as 25 percent across key product categories. These policy shocks triggered a re-configuration of trade flows, as firms sought to minimize tariff exposure through rerouting, regulatory arbitrage, and alternative delivery mechanisms.

The benchmark framework in the literature predicts substitution away from targeted suppliers and into untaxed partners (Anderson and Van Wincoop, 2003; Fajgelbaum et al., 2020). However, the adjustment margins exploited in practice extend beyond conventional trade reallocation. Empirical studies show that higher U.S. tariffs on Chinese goods generated sharp declines in direct U.S.–China trade while increasing exports from countries such as Vietnam, Thailand, Korea, and Mexico (Dang et al., 2023; Fajgelbaum et al., 2024; Nicita, 2019). Related work documents broader patterns of trade diversion and production relocation following tariff increases (Bown and Crowley, 2007; Flaaen et al., 2020).

At the same time, the rapid expansion of direct-to-consumer e-commerce created new opportunities to bypass tariffs through fragmented small-package shipments. Under the U.S. *de minimis* rule, imports valued at \$800 or less are exempt from customs duties, entry documentation, and inspection requirements. Although the exemption was not intended for commercial resellers or fragmented wholesale shipments, enforcement has remained limited. The scale of *de minimis* trade has therefore increased dramatically, rising from roughly 139 million parcels in 2015 to more than 1.36 billion shipments in 2024.² Entry Type 86 filings alone increased from approximately 122 million shipments in 2020 to nearly 950 million by 2024, with most arriving by air freight. This surge has strained customs enforcement capacity and complicated duty collection, product safety inspections, and intellectual property enforcement (Sutter and Sutherland, 2025).

These flows are largely absent from official trade statistics. U.S. import data based on Customs Value or Imports for Consumption do not systematically record shipments entering through *de minimis* channels. As a result, shifts toward fragmented parcel trade can simultaneously evade tariffs and measurement.

This paper asks how firms adjusted to U.S.–China trade war tariffs when they could circumvent duties through fragmented *de minimis* shipments or third-country rerouting.

²CBP (2024), “CBP Proposes New Rule to Strengthen Enforcement and Limit Duty Exemption for Low-Value Shipments,” <https://www.cbp.gov/newsroom/national-media-release/cbp-proposes-new-rule-strengthen-enforcement-and-limit-duty>.

In particular, the paper studies whether products with access to preexisting small-parcel shipment channels responded differently to tariff increases than products traded primarily through traditional customs channels. The central empirical challenge is that *de minimis* shipments are excluded from official U.S. trade statistics. Consequently, tariff avoidance through fragmented parcel trade appears in the data as a decline in recorded imports rather than an observable increase in low-value shipments.

The identification strategy therefore compares how tariff-exposed products with pre-trade war access to *de minimis* shipment channels respond relative to products without access to these channels. Products capable of shifting into fragmented parcel trade should exhibit larger declines in recorded imports following tariff increases, even conditional on similar tariff exposure. To implement this strategy, the analysis constructs a new shipment-based measure of *de minimis* eligibility using proprietary customs records obtained through a Freedom of Information Act request to U.S. Customs and Border Protection. The data cover all China-origin shipments valued below \$800 during the first week of December 2017, prior to the onset of the trade war. Cargo descriptions are mapped to HS product codes using text embeddings and validation procedures to identify products actively traded through fragmented parcel channels before tariff escalation. Products frequently observed in these shipments are classified as *de minimis* eligible.

This approach differs from existing work on *de minimis* trade. Fajgelbaum and Khandelwal (2024) study low-value imports using credit card and consumer transaction data, while Deng et al. (2024) proxy for *de minimis* eligibility using China’s Cross-Border E-Commerce export lists and infer evasion from discrepancies between reported Chinese exports and recorded U.S. imports. However, export-import gaps are inherently noisy because of timing discrepancies, valuation differences, and reporting inconsistencies, as emphasized in earlier work on mirror statistics and customs evasion (Fisman and Wei, 2004; Ferrantino et al., 2012). Other approaches, including applications of Benford’s law, have also been used to detect trade misreporting (Demir and Javorcik, 2020).

Rather than relying on policy-designated export categories or mirror-trade discrepancies, the classification in this paper is based on observed shipment behavior in pre-trade war customs records. The resulting measure captures products already entering through fragmented parcel channels and therefore most capable of exploiting the *de minimis* exemption once tariffs increased. These products are concentrated in consumer-oriented sectors such as apparel, electronics, toys, jewelry, and household goods, consistent with the rise of direct-to-consumer platforms such as Shein, Temu, and AliExpress (Couture et al., 2021).

The empirical analysis combines monthly HS10-level trade data from the U.S. Census Bureau with tariff schedules between 2017 and 2019. Tariff exposure is measured as the logarithmic change in one plus the ad valorem tariff rate, following Fajgelbaum et al. (2024),

Amiti et al. (2019), and Chor and Li (2024). The analysis excludes products first targeted under List 4A tariffs because these later tariff rounds overlap with substantial anticipatory adjustment and the onset of pandemic-era trade disruptions. Restricting the sample to pre-List 4A products isolates cleaner variation in tariff exposure before large-scale reorganization of shipping patterns.

The results show sharp declines in recorded U.S. imports from China following tariff increases, with significantly larger declines among products previously observed in *de minimis* shipment channels. The evidence is consistent with substitution into fragmented parcel trade that bypasses both tariffs and official measurement. At the same time, products exposed to *de minimis* shipment channels exhibit substantially weaker measured tariff incidence and pass-through, suggesting that a smaller share of trade activity remained within formal customs systems following tariff escalation. The paper also studies rerouting behavior through third countries using a bilateral flow-based measure adapted from Iyoha et al. (2024). For each bystander country and product, the rerouting share is defined as the minimum of the bystander’s exports to the United States and its imports from China, divided by total exports to the United States. This captures the share of trade flows potentially consistent with transshipment through an intermediary country.

Although the measure cannot fully distinguish transshipment from legitimate processing trade, it provides a transparent upper bound on rerouting intensity in tariffed sectors. Combining shipment-based *de minimis* classification with cross-country trade comparisons allows the analysis to distinguish fragmented parcel substitution from rerouting activity and to quantify how tariff circumvention reshaped observed trade flows.

The final section develops a structural framework of export choice under tariff frictions and shipment constraints. Exporters choose among four mutually exclusive strategies: fragmentation into *de minimis* parcels, direct shipment through standard customs channels, rerouting through a bystander country, or exit from the U.S. market altogether. Comparative statics show that both rerouting and *de minimis* fragmentation reduce the effective incidence of tariffs on targeted products, consistent with the empirical evidence.

The remainder of the paper is organized as follows. Section 2 provides background on *de minimis* regulations and the U.S.–China trade war. Section 3 outlines the data sources and construction of key variables. Section 4 presents the empirical strategy and benchmark regression specifications. Section 5 introduces the structural framework. Section 6 concludes.

2 Background: The U.S.–China Trade War and *De Minimis* Circumvention

The U.S.–China trade war formally began in 2018 following investigations under Section 301 of the Trade Act of 1974, which cited unfair trade practices by China, including forced technology transfer, intellectual property theft, and discriminatory licensing. The broader shift in U.S. trade policy, however, began earlier with safeguard tariffs on washing machines in January 2018 and national-security tariffs on steel and aluminum under Section 232 in March 2018. These measures were applied not only to China but also to major U.S. trade partners, including Canada, Mexico, and the European Union.

The subsequent escalation of Section 301 tariffs is summarized in Figure 1. In July 2018, the United States imposed a 25 percent tariff on \$34 billion in Chinese imports, followed one month later by an additional \$16 billion tranche. In September 2018, tariffs were extended to \$200 billion in imports at a 10 percent rate, which increased to 25 percent in May 2019. A final round covering roughly \$300 billion in remaining imports was announced in August 2019 and partially implemented through Lists 4A and 4B. By 2019, tariff measures covered approximately 72 percent of Chinese exports to the United States. China responded with retaliatory tariffs targeting U.S. exports such as agricultural products, automobiles, and energy goods. Although the Phase One Agreement in January 2020 temporarily de-escalated the conflict, most tariffs remained in place and continue under the current U.S. trade-policy framework.

Alongside the expansion of tariff coverage, the role of low-value parcel shipments grew sharply. Under Section 321 of the Tariff Act, imports valued at \$800 or less are eligible for entry under the U.S. *de minimis* exemption, which waives customs duties and substantially reduces documentation and inspection requirements for qualifying shipments. The threshold was increased from \$200 to \$800 in 2016, substantially lowering the administrative cost of fragmented parcel trade. Although the exemption was originally intended to simplify customs processing for low-value consumer shipments, firms increasingly used the provision to ship goods directly to consumers in small parcels rather than through traditional wholesale or containerized channels.

The scale of this expansion is shown in Figure 2. The number of *de minimis* shipments increased from roughly 411 million entries in 2018 to more than 1 billion by 2023, peaking at approximately 1.36 billion shipments in 2024 according to official CBP statistics. Entry Type 86 filings alone rose from roughly 122 million shipments in 2020 to nearly 950 million by 2024. These flows now account for a substantial share of U.S. consumer-goods imports and e-commerce transactions.

China overwhelmingly dominates this trade channel. Figure 3 shows that China-origin

shipments account for the majority of all *de minimis* entries, far exceeding volumes from other countries. The rapid growth of these flows coincided with the expansion of direct-to-consumer e-commerce platforms such as Shein, Temu, and AliExpress, which rely heavily on integrated parcel-shipping networks and air freight logistics. Chinese exporters increasingly used bonded warehouses, digital fulfillment systems, and cross-border e-commerce infrastructure to ship directly to U.S. consumers without relying on traditional intermediaries.

These developments created important enforcement and measurement challenges. Because *de minimis* shipments receive simplified customs treatment and reduced inspection, CBP processes millions of low-value parcels each day with limited product information and minimal verification capacity. This fragmentation complicates tariff collection, intellectual property enforcement, product safety inspections, and country-of-origin verification (Sutter and Sutherland, 2025). At the same time, *de minimis* entries are excluded from standard U.S. import statistics published by Census and Customs. As trade increasingly shifts toward fragmented parcel channels, official trade statistics become less informative about the true incidence of tariffs and the composition of imports entering the United States.

Because public trade statistics do not systematically record *de minimis* shipments, identifying products capable of shifting into fragmented parcel channels requires external shipment-level information. The analysis therefore uses proprietary CBP shipment records covering China-origin shipments valued below \$800 prior to the trade war. Products frequently observed in these shipments are classified as *de minimis* exposed.

The combination of rising tariff exposure and expanding parcel-shipping infrastructure created strong incentives for firms to substitute toward unobserved *de minimis* trade during the U.S.–China trade war. As the U.S. economy becomes increasingly reliant on fragmented, high-frequency e-commerce shipments, the limitations of conventional trade-policy tools become increasingly apparent.

3 Data

The empirical analysis combines trade datasets spanning 2017–2019 at both monthly and quarterly frequencies. Baseline import specifications use monthly HS10-level U.S. Census data, while rerouting specifications use quarterly HS6-level bilateral trade flows harmonized under the 2012 revision of the Harmonized System. Quarterly aggregation is used in the rerouting analysis to reduce noise in intermediary-country trade flows. The data are organized at the country–product–time level.

3.1 Trade Data Sources and Coverage

The empirical analysis combines monthly U.S. import data at the HS10 level with quarterly bilateral international trade flows at the HS6 level spanning 2017–2019. U.S. import and tariff revenue data are obtained from the U.S. Census Bureau and include customs value, import quantity, calculated duties, and effective duty measures at the HS10–country–month level. Bilateral trade flows for China and third-country partners are obtained from the United Nations Comtrade Database, which compiles official import and export statistics reported by national customs authorities under the 2012 revision of the Harmonized System. Quarterly aggregation is used for the bilateral trade-flow analysis to reduce volatility in intermediary-country rerouting measures.

All trade values use the primary valuation reported by the source. UN Comtrade records imports on a cost–insurance–freight (CIF) basis and exports on a free-on-board (FOB) basis, while U.S. Census “General Imports” report customs value including duties and charges.³ Bilateral trade flows are harmonized to the HS6 level when combining across reporting systems, while baseline import and tariff specifications retain the underlying HS10 Census classification.

The combined dataset includes U.S. imports from China and from third-country bystanders, Chinese exports to the United States and to bystander countries, and bystander imports from China and exports to the United States. Together, these flows capture both direct and indirect trade linkages and allow for the identification of rerouting behavior following tariff escalation.

Additional pre-trade-war Census data from 2015–2017 are used to construct product-level shipment characteristics unaffected by tariff distortions. These data include customs value, CIF value, and mode-of-transport information by HS10 product, which are used to estimate average transport costs, air-shipment intensity, and other shipment characteristics associated with *de minimis* eligibility and parcelization.

Quarterly HS6 bilateral trade flows are used in the rerouting analysis to reduce volatility in intermediary-country trade patterns and short-run fluctuations in monthly trade reporting. These data are used to construct measures of indirect trade exposure and potential transshipment through intermediary economies during the trade war period.

³UN Comtrade reports imports on a CIF basis and exports on an FOB basis, while U.S. Census “General Imports” provide customs value including duties and charges and are the closest comparable measure. Results are robust to alternative scaling adjustments that account for freight and insurance margins.

3.2 Tariff Data

Product-level tariff rates are constructed using the Peterson Institute for International Economics (PIIE) Section 301 tariff schedules compiled by Bown (2022), which provide detailed HTS-level coverage of all major U.S.–China trade war tariff actions. These data include the successive Section 301 tariff lists implemented between 2018 and 2019, including Lists 1, 2, 3, and 4A, as well as safeguard tariffs on washing machines and solar panels and Section 232 tariffs on steel and aluminum. Tariff schedules are matched to U.S. import data at the HTS10 level and aggregated to HS6 where necessary for consistency with international trade data.

The analysis focuses on the initial phase of the trade war between 2017 and 2019 and excludes products first treated under List 4A tariffs because these products were exposed after firms had already begun adjusting sourcing and shipment behavior. Restricting the sample to earlier tariff waves reduces concerns related to anticipatory effects, supply-chain reorganization, and overlap with subsequent COVID-era trade disruptions.

Applied tariff exposure is measured using the ad valorem Section 301 tariff rate. These duties are additive to preexisting most-favored-nation (MFN) tariffs, which are obtained from the World Trade Organization (WTO) tariff schedules. The average MFN tariff on Chinese imports before the trade war was approximately 3 percent, while Section 301 duties increased effective tariff rates substantially for treated products.

3.3 De Minimis Classification from Shipment Data

To identify products likely to enter the United States through fragmented small-package shipment channels, the analysis uses shipment-level records obtained through a Freedom of Information Act request to U.S. Customs and Border Protection (CBP). The data contain transaction-level observations for low-value shipments, including detailed cargo descriptions and declared shipment values.

Products are classified as *de minimis* exposed by mapping shipment descriptions to HS6 product codes using text embeddings and validation procedures described in Appendix A. Rather than relying on policy-based eligibility or external e-commerce classifications, the measure identifies products that were actively traded through low-value parcel channels before the onset of the trade war.

The baseline classification uses China-origin shipments arriving during the first week of December 2017 with declared values below the \$800 *de minimis* threshold. A product is classified as *de minimis* if it is observed in at least 25 low-value shipments during the sample week. This procedure identifies 736 HS6 products with pre-trade war exposure to

fragmented parcel trade.⁴

The resulting classification is heavily concentrated in consumer-oriented sectors commonly associated with direct-to-consumer e-commerce. Table 1 shows that the highest shares of *de minimis* products occur in apparel, footwear, toys, leather goods, jewelry, and miscellaneous retail categories. More than 80 percent of knitted apparel HS6 products and nearly 70 percent of non-knitted apparel products are classified as *de minimis* exposed. By contrast, machinery and industrial inputs exhibit substantially lower exposure rates.

Table 2 further shows that *de minimis* products differ systematically from non-*de minimis* goods along several shipment characteristics. Products classified as *de minimis* exposed are significantly more likely to be shipped by air and substantially more concentrated in final consumption categories, while non-*de minimis* products are disproportionately intermediate goods shipped through traditional vessel-based trade channels. These patterns are consistent with fragmented e-commerce logistics and direct-to-consumer shipment models.

Figure 4 examines the timing of tariff exposure across product groups. Non-*de minimis* products were disproportionately targeted in earlier tariff waves, particularly List 3, while *de minimis* products were more heavily concentrated in later rounds such as List 4A. Figure 5 shows that this timing difference reflects the composition of tariff waves themselves: earlier rounds primarily targeted machinery and industrial inputs, whereas later rounds increasingly covered textiles, apparel, and consumer goods that were more compatible with small-package shipment channels.

These compositional differences are important for the empirical design. Because *de minimis* products were systematically exposed later in the trade war, the baseline analysis focuses on the earlier tariff waves and excludes products first treated under List 4A. This restriction reduces concerns related to anticipatory adjustment, evolving shipment behavior, and overlap with pandemic-era trade disruptions.

3.4 Rerouting Measure and Summary Statistics

To measure potential transshipment through intermediary economies, the analysis constructs a product-level rerouting share following Iyoha et al. (2024). For each product p , bystander country c , and quarter t , the rerouting share is defined as:

$$L_{pct} = \frac{\min\{x_{pct}^{US}, m_{pct}^{CHN}\}}{x_{pct}^{US}},$$

where x_{pct}^{US} denotes exports from bystander country c to the United States and m_{pct}^{CHN}

⁴Appendix A provides additional details on shipment coverage, embedding confidence, validation procedures, and alternative classification approaches.

denotes imports from China into the bystander country. The measure captures the maximum share of exports to the United States that could plausibly be sourced from China.

Figure 6 reports the distribution of rerouting shares across bystander-country-product observations in 2018Q1. The distribution is highly dispersed, with substantial mass near zero indicating no observable rerouting, alongside a smaller concentration near one consistent with intensive transshipment activity. Intermediate values reflect partial rerouting and mixed sourcing patterns.

Although the measure cannot fully distinguish transshipment from legitimate processing trade or production relocation, it provides a transparent upper-bound indicator of indirect trade exposure during the tariff period. Combined with the shipment-based *de minimis* classification, the rerouting measure helps distinguish between tariff avoidance through fragmented parcel trade and tariff avoidance through third-country intermediaries.

3.5 Descriptive Patterns

Table 3 reports average trade flows before and after the introduction of Section 301 tariffs in July 2018. U.S. imports from China decline following tariff escalation for both product groups, while imports from bystander countries increase over the same period, consistent with partial trade diversion and rerouting through intermediary economies. The increase in bystander trade flows is somewhat larger for non-*de minimis* products, consistent with heavier and intermediate goods being more likely to adjust through traditional rerouting channels rather than fragmented parcel shipments.

Table 4 summarizes the distribution of tariff exposure across products. Average tariff increases rise sharply after July 2018, reaching roughly 10 percentage points in the post-period. Although overall tariff exposure is similar across *de minimis* and non-*de minimis* products, Figure 4 shows that the timing of exposure differs systematically across groups. Non-*de minimis* products were disproportionately targeted in earlier tariff waves, particularly List 3, while *de minimis* products were more heavily exposed in later rounds such as List 4A.

These patterns are consistent with the underlying composition of the shipment-based *de minimis* classification. As shown previously, *de minimis* products are concentrated in apparel, footwear, toys, jewelry, and other consumer-oriented categories commonly associated with direct-to-consumer parcel trade. In contrast, non-*de minimis* goods are more heavily concentrated in machinery, industrial inputs, and intermediate products traded through conventional shipping channels.

Taken together, the descriptive evidence highlights substantial heterogeneity in how products were exposed to and adjusted during the trade war period. Products linked to fragmented parcel trade experienced different timing of tariff exposure and different patterns of

trade adjustment than products primarily traded through traditional customs channels.

4 Empirical Framework and Findings

Trade flows adjusted to the U.S.–China trade war along multiple margins, including direct reductions in bilateral imports, rerouting through intermediary economies, and changes in tariff incidence within the formal customs system. This section quantifies these adjustment channels using monthly HS10- and HS6-level trade data matched to product-level tariff exposure between January 2017 and August 2019.

The analysis excludes products first targeted under List 4A tariffs in order to focus on the earlier phase of the trade war before substantial anticipatory adjustment and pandemic-era disruptions. The specifications compare responses across products classified as *de minimis* exposed and non-*de minimis* using the shipment-based CBP classification developed in Section 3.

4.1 Model 1: Direct Impact on U.S. Imports from China

This model estimates how tariff exposure affected recorded U.S. imports from China across products with different exposure to *de minimis* shipment channels. The baseline specification is estimated at the HS10-month level:

$$\log(y_{it}) = \beta_1 \log(1 + \tau_{it}) + \beta_2 [DeMinimis_i \times \log(1 + \tau_{it})] + \gamma_i + \delta_t + \varepsilon_{it},$$

where y_{it} denotes U.S. import outcomes for product i in month t , including import value, import quantity, unit value, and tariff-inclusive prices. The variable τ_{it} denotes the applied MFN plus Section 301 tariff rate, while $DeMinimis_i$ indicates products classified as *de minimis* exposed using the CBP shipment-based measure. All specifications include HS10 and month fixed effects, and standard errors are clustered at the HS8 level.

The coefficient β_1 captures how tariff exposure affects products primarily traded through traditional customs channels, while β_2 measures whether products previously observed in low-value parcel shipments respond differently. A negative interaction term implies that products with access to *de minimis* shipment channels experience larger declines in recorded imports following tariff increases.

The empirical design compares tariff responses across products with different pre-trade war exposure to fragmented parcel trade. The identifying assumption is that, absent the trade war, import trends for *de minimis* and non-*de minimis* products would have evolved similarly conditional on product and time fixed effects. Under this assumption, differential

declines in recorded imports capture differences in how firms adjusted to tariff exposure across shipment channels.

Table 5 reports the baseline estimates. Tariff increases generate large declines in both import values and import quantities overall. The interaction term between tariff exposure and *de minimis* status is negative and statistically significant for import values, indicating that products with access to low-value shipment channels experienced larger declines in recorded imports following tariff escalation.

The estimated magnitudes are economically substantial. For non-*de minimis* products, a 25 percentage point tariff increase implies approximately a 33 percent decline in import values. For *de minimis* products, the combined coefficients imply a decline of roughly 39 percent. One interpretation is that firms importing consumer-oriented goods were better able to reorganize shipments into fragmented parcel channels once tariffs increased, causing a larger share of trade activity to disappear from standard customs statistics.

Columns (3) and (4) show that tariff pass-through also differs across product groups. Tariff exposure raises observed unit values and tariff-inclusive prices overall, consistent with conventional tariff incidence. However, the interaction terms are negative and statistically significant for *de minimis* products, implying weaker measured pass-through among goods more exposed to small-package shipment channels. This pattern is consistent with partial avoidance of formal tariff collection.

Figure 7 presents dynamic treatment effects using the staggered-adoption event-study estimator of Sun and Abraham (2021). The event-study specification estimates month-by-month responses relative to the period immediately preceding tariff exposure while accounting for heterogeneous treatment timing across products.

Across both product groups, pre-treatment coefficients remain relatively close to zero, providing little evidence of differential pre-trends prior to tariff implementation. Following treatment, import values and quantities decline sharply for both groups, although the post-treatment decline is larger and more volatile for *de minimis* products. Non-*de minimis* goods exhibit clearer increases in unit values and tariff-inclusive prices after treatment, consistent with standard tariff pass-through. In contrast, price responses for *de minimis* products are weaker and less tightly linked to the timing of tariff implementation.

These dynamic patterns reinforce the baseline regression results. Non-*de minimis* products primarily adjust through conventional price and quantity margins within formal customs channels, whereas *de minimis* products exhibit larger declines in recorded imports alongside weaker measured price pass-through.

Table 6 examines robustness to alternative proxies for *de minimis* suitability, including China’s CBEC list, products with unit values below \$800, consumption-good classifications, and products commonly shipped by air. The strongest and most consistent effects emerge us-

ing the shipment-based CBP classification and related parcel-trade proxies, whereas broader consumption-based classifications generate substantially weaker and less stable estimates. These results reinforce the interpretation that the baseline measure captures actual exposure to fragmented shipment channels rather than broad product composition differences alone.

4.2 Model 2: Tariff Incidence and Revenue Collection

This model examines whether tariff exposure translated into observable tariff payments within the formal customs system, or whether products exposed to fragmented shipment channels exhibited weaker measured tariff incidence. The analysis focuses on monthly U.S. imports from China at the HS6-product level.

The baseline specification is:

$$\log(y_{it}) = \beta_1 \log(1 + \tau_{it}) + \beta_2 [DeMinimis_i \times \log(1 + \tau_{it})] + \gamma_i + \delta_t + \varepsilon_{it},$$

where y_{it} alternatively denotes log duties paid and the log effective tariff rate, defined as:

$$\log \left(\frac{\text{Duties Paid}_{it}}{\text{Import Value}_{it}} \right).$$

The variable τ_{it} denotes the applied MFN plus Section 301 tariff rate, while $DeMinimis_i$ indicates products classified as *de minimis* exposed using the CBP shipment-based measure. All specifications include HS6 and month fixed effects, and standard errors are clustered at the HS6 level.

The coefficient β_1 captures how tariff exposure affects products primarily imported through traditional customs channels, while β_2 measures whether products more exposed to fragmented parcel trade exhibit different patterns of tariff collection. A negative interaction term implies that tariff exposure translates into weaker observed duty collection among products more compatible with *de minimis* shipment channels.

Table 7 reports the baseline estimates. Tariff exposure strongly increases both calculated duties paid and effective tariff rates overall. The baseline coefficients are large and highly statistically significant across all specifications, consistent with substantial tariff incidence on formally recorded Chinese imports.

However, the interaction terms between tariff exposure and *de minimis* status are negative and statistically significant for both outcomes. Products classified as *de minimis* exposed exhibit substantially weaker growth in measured duties paid and effective tariff rates following tariff increases. These estimates imply that tariff incidence was concentrated disproportionately among products continuing to enter through traditional customs channels.

The estimated magnitudes are economically meaningful. In column (2), the interaction coefficient implies that products exposed to fragmented shipment channels experienced substantially smaller increases in observed duties paid following tariff escalation. Column (4) shows an even larger negative interaction effect for effective tariff rates, indicating that the share of import value collected as duties rose significantly less for *de minimis* products than for comparable non-*de minimis* goods.

These findings are consistent with the broader adjustment patterns documented in the previous section. Products more compatible with fragmented parcel trade experienced larger declines in recorded imports alongside weaker measured tariff collection, suggesting that firms were better able to reorganize trade flows outside traditional customs channels. In contrast, non-*de minimis* products remained more exposed to formal tariff collection and conventional customs enforcement.

Taken together, the results indicate that the fiscal incidence of Section 301 tariffs fell disproportionately on products continuing to operate within standard import channels, while products suited to low-value shipment fragmentation exhibited substantially weaker measured tariff collection.

4.3 Model 3: Rerouting Through Third Countries

This model examines whether tariff exposure increased rerouting through intermediary economies following the imposition of Section 301 tariffs. The analysis uses quarterly HS6-partner observations and measures indirect trade along the China $\rightarrow$ bystander country $\rightarrow$ United States pathway.

The baseline specification is:

$$\text{Reroute}_{ipt} = \beta_1 \tau_{it} + \beta_2 [\text{DeMinimis}_i \times \tau_{it}] + \alpha_{ip} + \delta_t + \gamma_{pt} + \varepsilon_{ipt},$$

where Reroute_{ipt} denotes the rerouting-share measure developed by Iyoha et al. (2024):

$$\text{Reroute}_{ipt} = \frac{\min \left(X_{it}^{CHN \rightarrow p}, M_{it}^{US \leftarrow p} \right)}{M_{it}^{US \leftarrow p}},$$

with $X_{it}^{CHN \rightarrow p}$ denoting Chinese exports to bystander country p and $M_{it}^{US \leftarrow p}$ denoting U.S. imports from bystander country p for product i in quarter t . The variable τ_{it} denotes the applied MFN plus Section 301 tariff rate, while DeMinimis_i indicates products classified as *de minimis* exposed using the CBP shipment-based measure. All specifications include HS6 $\times$ partner and partner $\times$ quarter fixed effects, and standard errors are clustered at the HS6 level.

The coefficient β_1 captures how tariff exposure affects rerouting among products primarily traded through traditional customs channels, while β_2 measures whether products more exposed to fragmented parcel trade respond differently. A negative interaction term implies that products suited to low-value shipment channels experience weaker rerouting responses following tariff escalation.

The empirical design compares rerouting responses across products with different pre-trade war exposure to fragmented parcel trade. The identifying assumption is that, absent the trade war, rerouting trends for *de minimis* and non-*de minimis* products would have evolved similarly conditional on product-partner and time fixed effects.

Table 8 reports the baseline estimates. Columns (1) and (2) use the rerouting-share measure from Iyoha et al. (2024), while columns (3) and (4) use a relative trade-flow measure comparing Chinese exports to bystander countries with U.S. imports from those same countries.

Across both measures, tariff exposure significantly increases rerouting through intermediary economies. In column (2), the interaction term between tariff exposure and *de minimis* status is negative and statistically significant, implying substantially weaker rerouting responses among products more compatible with fragmented parcel shipment channels.

The estimated magnitudes are economically meaningful. In the baseline rerouting-share specification, a 25 percentage point tariff increase raises the rerouting share for non-*de minimis* products by roughly one percentage point.⁵ The negative interaction effect largely offsets this increase for *de minimis* products, implying little additional rerouting among goods more easily shipped through low-value parcel channels.

Columns (3) and (4) show similar patterns using the alternative relative trade-flow specification. Tariff exposure increases indirect trade flows through intermediary economies overall, while the interaction term indicates substantially weaker responses for *de minimis* products. The consistency across both rerouting measures suggests that the results are not driven by the construction of any single dependent variable.

Table 9 examines robustness to alternative proxies for *de minimis* suitability, including China’s CBEC list, products with unit values below \$800, consumption-good classifications, and products commonly shipped by air. The strongest negative interaction effects emerge using shipment-based and consumer-oriented parcel-trade proxies, while broader classifications produce weaker or statistically insignificant estimates.

Table 10 further examines heterogeneity across product characteristics associated with shipment fragmentation and trade costs. Rerouting responses are weakest among products least commonly shipped by air, consistent with heavier or bulkier goods being less suited

⁵Using column (2), a 25 percentage point tariff increase corresponds to an increase of approximately $0.25 \times 0.038 \approx 0.01$ in the rerouting share.

to fragmented parcel delivery. Taken together, these results suggest that rerouting through intermediary economies was concentrated primarily among products continuing to rely on traditional customs and shipping channels, whereas products compatible with low-value parcel trade adjusted through different margins.

4.4 Interpretation of Results

Taken together, the three sets of results point to a systematic reorganization of trade flows in response to the U.S.–China trade war. Firms adjusted along multiple margins simultaneously, with the form of adjustment depending heavily on whether products could be fragmented into low-value parcel shipments.

The direct-import results show that tariff exposure reduced recorded U.S. imports from China overall, but the decline was substantially larger for products previously observed in *de minimis* shipment channels. At the same time, these products exhibited weaker increases in observed unit values and tariff-inclusive prices, suggesting less conventional tariff pass-through within formal customs data.

The tariff-incidence results reinforce this interpretation. Although tariff exposure sharply increased calculated duties and effective tariff rates overall, the increase was significantly smaller for products classified as *de minimis* exposed. This pattern indicates that products more compatible with fragmented parcel trade generated substantially weaker measured tariff collection following tariff escalation.

By contrast, the rerouting results show that indirect trade through intermediary economies increased most strongly for products continuing to rely on traditional shipping and customs channels. Non-*de minimis* products experienced economically meaningful increases in rerouting shares following tariff exposure, whereas products more exposed to low-value shipment channels exhibited substantially weaker rerouting responses.

Taken together, these findings suggest that firms adjusted differently depending on the logistical characteristics of the product. Products suited to fragmented parcel trade appear to have shifted toward shipment channels less visible in standard import statistics and less exposed to formal tariff collection. In contrast, products less compatible with small-package shipment continued to move through traditional customs systems, where adjustment occurred through rerouting, tariff pass-through, and conventional trade diversion.

More broadly, the results imply that standard trade statistics may substantially understate true import activity during periods of elevated trade barriers when firms can reorganize transactions across shipment channels. The effectiveness and fiscal incidence of tariff policy therefore depend not only on tariff rates themselves, but also on the ability of firms to fragment shipments and shift trade into alternative customs regimes.

5 A Structural Framework for Trade Policy and Evaluation

The preceding empirical analysis uses a difference-in-differences approach to estimate how firms responded to U.S. trade war tariffs by rerouting shipments or exploiting the *de minimis* exemption. While this strategy identifies reduced-form effects, it does not fully capture the underlying economic trade-offs that shape firm decisions across multiple margins.

To complement the empirical findings, this section introduces a structural model that formalizes firms' export channel choices under trade policy shocks. The framework allows firms to endogenously select among multiple discrete shipment strategies and enables counterfactual simulations of trade reallocation. While the demand environment follows a standard CES structure, the model extends prior approaches by incorporating product-specific shipment costs, *de minimis* eligibility, the option to reroute goods through third countries, and the option value of fast shipment emphasized by Hummels and Schaur (2010). Firms choose among four mutually exclusive export channels: direct non-*de minimis* shipment to the United States, direct *de minimis* shipment to the United States, rerouting through a bystander country, and no export. These choices depend on tariff exposure, shipment size, product-specific transport frictions, and the value of rapid shipment under demand uncertainty.

This structural approach serves two purposes. First, it sharpens the interpretation of the empirical estimates by linking observed trade responses to unobserved cost and regulatory parameters. Second, it provides a foundation for policy counterfactuals, such as evaluating how global trade patterns would have differed if the United States had eliminated the *de minimis* exemption more broadly.

5.1 Environment

The world economy consists of a set $\mathcal{I}$ of countries, indexed by i for exporters. Individual products are indexed by m . Each product m is differentiated by its country of origin i , forming a distinct variety im . Competitive producers in country i sell variety im at domestic price p_{im} .

5.1.1 Supply and Firm Channel Choices

Firms are assumed to be homogeneous within each product m and choose the export channel k that yields the highest profit. For any given channel, profit is defined as the revenue from selling one unit in the U.S. market minus the total costs incurred when using

that channel. The choice of channel depends on product-specific characteristics, including shipment size s_m , transport suitability ϕ_m , and whether the product qualifies for *de minimis* treatment.

Direct De Minimis (Channel i): *De minimis* entry allows firms to fragment shipments into low-value parcels that enter without tariffs. Because these shipments are typically transported by air and delivered directly to consumers, this channel also provides speed and inventory flexibility relative to bulk shipment. Following Hummels and Schaur (2010), fast transport has option value when demand is uncertain because firms can delay shipment decisions until demand conditions are realized, reducing inventory risk and improving responsiveness to market conditions.

The profit for this channel, for quantity Q , is

$$\pi_m^{\text{deMin}}(Q) = p_m^{US}Q - c_mQ - t_m^{\text{deMin}}(s_m, \phi_m)Q - \phi_m^{\text{pkg}}(s_m, \phi_m)Q + \Omega_m(\sigma_m^D, A_m)Q, \quad (1)$$

where c_m is the marginal cost of production in China for product m . The term $\Omega_m(\sigma_m^D, A_m)$ captures the option value of fast shipment following Hummels and Schaur (2010). The option value is increasing in product-level demand volatility σ_m^D and realized demand conditions A_m , reflecting the value of rapid delivery and inventory flexibility for products facing uncertain or fluctuating demand.

The *de minimis* shipment cost, $t_m^{\text{deMin}}(s_m, \phi_m)$, is linear in shipment size s_m , with the slope adjusted by the product's air-friendliness ϕ_m :

$$t_m^{\text{deMin}}(s_m, \phi_m) = \alpha_{\text{air}} \cdot s_m \cdot \exp(-\beta_{\text{air}}\phi_m). \quad (2)$$

The parameter α_{air} represents the baseline per-unit shipping cost for non-air-friendly goods, while β_{air} governs the sensitivity of shipping costs to air-friendliness, reducing the cost for products that are easier to ship by air. This specification captures the high per-unit cost characteristic of air freight, often associated with *de minimis* shipments, and reflects that goods with higher ϕ_m incur lower shipping costs.

The packaging or friction cost per unit, $\phi_m^{\text{pkg}}(s_m, \phi_m)$, is modeled as

$$\phi_m^{\text{pkg}}(s_m, \phi_m) = \phi_0 + \phi_1 \cdot s_m^\gamma \cdot \exp(\beta_{\text{pkg}}\phi_m). \quad (3)$$

Here, ϕ_0 represents a fixed handling cost and $\phi_1 \cdot s_m^\gamma$ captures the size-dependent complexity of packaging. If $\gamma > 1$, packaging costs increase disproportionately with shipment size. The parameter ϕ_1 captures how packaging costs escalate with size, while β_{pkg} accounts for heterogeneity in packaging difficulty across goods.

Direct Non-De Minimis (Channel ii): This channel requires firms to pay the full

U.S. tariff, τ_m^{US} , and is typically associated with larger bulk shipments transported through standard customs channels. The profit for this channel is

$$\pi_m^{\text{dir}}(Q) = p_m^{\text{US}}Q - c_mQ - (p_m^{\text{US}} \cdot \tau_m^{\text{US}})Q - t_m^{\text{dir}}(s_m, \phi_m)Q. \quad (4)$$

The direct shipment cost per unit, $t_m^{\text{dir}}(s_m, \phi_m)$, is assumed to follow the functional form

$$t_m^{\text{dir}}(s_m, \phi_m) = \alpha_{\text{bulk}} \cdot s_m^\theta \cdot \exp(\beta_{\text{bulk}}\phi_m), \quad (5)$$

with the condition that $\theta < 1$. This functional form captures economies of scale, implying that the per-unit cost of shipment decreases as shipment size increases. The $\exp(\beta_{\text{bulk}}\phi_m)$ term allows products better suited for bulk shipping to incur lower per-unit transport costs.

Rerouting through a Bystander (Channel iii): This strategy entails transshipment through an intermediate bystander country b . Consequently, it incurs tariffs in both the bystander country ($\tau_m^{\text{CN},b}$) and the United States ($\tau_m^{b,\text{US}}$), in addition to combined transport costs for the entire route. The profit for this channel is expressed as

$$\pi_m^{\text{reroute}}(Q) = p_m^{\text{US}}Q - c_mQ - (p_m^{\text{US}} \cdot \tau_m^{\text{CN},b})Q - (p_m^{\text{US}} \cdot \tau_m^{b,\text{US}})Q - t_m^{\text{reroute}}(s_m, \phi_m)Q - f_m^{\text{reroute}}. \quad (6)$$

The combined shipment cost per unit for this rerouting path, $t_m^{\text{reroute}}(s_m, \phi_m)$, is the sum of the costs for each leg:

$$t_m^{\text{reroute}}(s_m, \phi_m) = t_m^{\text{CN} \rightarrow b}(s_m, \phi_m) + t_m^{b \rightarrow \text{US}}(s_m, \phi_m). \quad (7)$$

A fixed cost for rerouting, f_m^{reroute} , captures non-convexities such as establishing distribution networks or legal entities in a bystander country. This fixed cost is decreasing in the product's market size:

$$f_m^{\text{reroute}} = \alpha_f \cdot \text{SIZE}_{md}^{-1}. \quad (8)$$

No Export (Channel iv): If none of the available channels generate positive profit, the firm chooses not to export:

$$\pi_m^{\text{none}}(Q) = 0. \quad (9)$$

For each product m , the firm chooses the feasible channel that maximizes profits:

$$x_m^* = \arg \max_{k \in \{\text{deMin}, \text{dir}, \text{reroute}, \text{none}\}} \pi_m^{(k)}(Q). \quad (10)$$

Regarding aggregate supply, the total sales of product m from country i , denoted X_{im} , are determined by its domestic price p_{im} and an inverse supply elasticity b_i^j . This relationship

is expressed as

$$X_{im} = A_i^j \cdot p_{im}^{\frac{1}{b_i^j}} \cdot Z_{im}, \quad (11)$$

where b_i^j represents the inverse supply elasticity, defined as the elasticity of price with respect to total sales. A_i^j is an endogenous country-sector component that captures factor and input prices, while Z_{im} is an exogenous cost shifter.

5.1.2 Demand

In the partial equilibrium framework, the United States is the sole destination market. Consumers allocate fixed expenditure across imported varieties of a product m , with varieties differentiated by country of origin. Preferences are represented by a constant elasticity of substitution (CES) utility function.

Let p_{im} denote the delivered price of variety im —product m from origin country i . Delivered prices depend on the firm's chosen export channel and are given by:

$$p_{im} = T_{im} \cdot \tau_{im} \cdot p_{im}^s, \quad (12)$$

where p_{im}^s is the producer price in the origin country, τ_{im} captures iceberg trade costs, and T_{im} represents one plus the effective ad valorem tariff applied to imports from i into the United States.

Different channels imply different combinations of tariffs and trade costs. Under the *de minimis* exemption, the tariff component satisfies $T_{im} = 1$, reflecting zero tariff liability, and τ_{im} corresponds to direct parcel shipment to the United States. For direct non-*de minimis* shipment, T_{im} reflects the full tariff imposed on origin country i , while τ_{im} reflects bulk shipment costs. For rerouted goods, T_{im} reflects cumulative tariffs across both segments of the rerouted path, while τ_{im} incorporates total transport costs across the two legs.

Total expenditure E_m on product m is fixed. Given delivered prices, demand follows the CES allocation rule:

$$q_{im} = \left(\frac{p_{im}}{P_m} \right)^{-\sigma} E_m, \quad (13)$$

where σ is the elasticity of substitution across varieties and P_m is the CES price index:

$$P_m = \left(\sum_i p_{im}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \quad (14)$$

5.1.3 Partial Equilibrium and Exporter Problem

Firms endogenously select an export channel for each product-destination pair in order to maximize expected profits. The choice of channel affects both delivered prices and the fixed and variable costs of exporting. Channels differ in terms of tariff exposure, shipment cost, eligibility constraints, and rerouting frictions.

Because the framework is partial equilibrium, prices and aggregate expenditure are treated as exogenous. The model therefore isolates the effects of trade policy on exporter behavior while abstracting from general equilibrium adjustments in wages, factor markets, and aggregate income.

5.2 Comparative Statics

This section characterizes how changes in tariffs, shipment size, and product attributes alter the relative profitability of alternative export channels within the structural framework.

For a product m with shipment size s_m and transport suitability ϕ_m , the net per-unit generalized costs of each feasible channel are defined as

$$G_m^{\text{deMin}}(s_m, \phi_m, \sigma_m^D, A_m) \equiv t_m^{\text{deMin}}(s_m, \phi_m) + \phi_m^{\text{pkg}}(s_m, \phi_m) - \Omega_m(\sigma_m^D, A_m), \quad (15)$$

$$G_m^{\text{dir}}(s_m, \phi_m) \equiv p_m^{\text{US}} \tau_m^{\text{US}} + t_m^{\text{dir}}(s_m, \phi_m), \quad (16)$$

$$G_m^{\text{re}}(Q, s_m, \phi_m) \equiv p_m^{\text{US}} (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}}) + t_m^{\text{reroute}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q}. \quad (17)$$

Channel choice is equivalent to minimizing generalized cost over the feasible set of export strategies. The inclusion of $\Omega_m(\sigma_m^D, A_m)$ implies that the attractiveness of *de minimis* shipment depends not only on tariff avoidance, but also on the value of rapid delivery and inventory flexibility under demand uncertainty.

Pairwise dominance conditions define the regions in which each channel is preferred. *De minimis* dominates direct non-*de minimis* shipment when

$$t_m^{\text{deMin}}(s_m, \phi_m) + \phi_m^{\text{pkg}}(s_m, \phi_m) - \Omega_m(\sigma_m^D, A_m) < p_m^{\text{US}} \tau_m^{\text{US}} + t_m^{\text{dir}}(s_m, \phi_m), \quad (18)$$

and dominates rerouting when

$$t_m^{\text{deMin}}(s_m, \phi_m) + \phi_m^{\text{pkg}}(s_m, \phi_m) - \Omega_m(\sigma_m^D, A_m) < p_m^{\text{US}} (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}}) + t_m^{\text{reroute}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q}. \quad (19)$$

Rerouting dominates direct non-*de minimis* shipment when

$$p_m^{\text{US}} (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}}) + t_m^{\text{reroute}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q} < p_m^{\text{US}} \tau_m^{\text{US}} + t_m^{\text{dir}}(s_m, \phi_m). \quad (20)$$

These inequalities define the loci separating the *de minimis*, direct, and rerouted regimes in the space of tariffs, shipment size, transport suitability, and operating scale.

Holding (s_m, ϕ_m, Q) fixed,

$$\frac{\partial}{\partial \tau_m^{\text{US}}} (G_m^{\text{dir}} - G_m^{\text{deMin}}) = \frac{\partial}{\partial \tau_m^{\text{US}}} (G_m^{\text{dir}} - G_m^{\text{re}}) = p_m^{\text{US}} > 0,$$

so increases in U.S. tariffs on Chinese goods expand the set of products for which *de minimis* shipment or rerouting dominates direct shipment.

Similarly,

$$\frac{\partial}{\partial \tau_m^{b,\text{US}}} (G_m^{\text{re}} - G_m^{\text{dir}}) = p_m^{\text{US}} > 0,$$

implying that higher tariffs on imports from bystander countries reduce the attractiveness of rerouting.

The role of shipment size and transport suitability follows directly from the cost functions. Under bulk economies of scale ($\theta < 1$) and convex packaging costs ($\gamma \geq 1$), there exists a shipment threshold $s_m^*(\phi_m)$ such that sufficiently small and air-suitable products favor *de minimis* shipment whenever eligibility holds. Greater air-friendliness lowers *de minimis* shipment costs and increases the relative value of rapid shipment under demand uncertainty, while greater bulk suitability lowers direct-shipping costs.

Scale affects the profitability of rerouting through the fixed-cost term. A cutoff scale

$$Q_m^* = \frac{f_m^{\text{reroute}}}{p_m^{\text{US}} (\tau_m^{\text{US}} - \tau_m^{\text{CN},b} - \tau_m^{b,\text{US}}) + (t_m^{\text{dir}} - t_m^{\text{reroute}})}$$

exists such that larger operating scale increases the profitability of rerouting relative to direct shipment.

These conditions yield clear policy and logistics implications. Increases in Section 301 tariffs enlarge the regions where *de minimis* shipment and rerouting become optimal, while stricter enforcement or higher packaging costs reduce the profitability of fragmented parcel trade and shift activity back toward formal customs channels. Higher tariffs on bystander economies shrink the rerouting region and partially restore direct trade. Taken together, the model predicts a sharper contraction in recorded non-*de minimis* China-U.S. trade than in *de minimis*-eligible products, greater expansion of rerouting for bulk-suited goods, and a

widening gap between observed imports and underlying trade activity as firms increasingly rely on fragmented shipment channels.

6 Calibration Framework (In Progress)

This paper simulates firm responses to trade war policies by modeling the profit-maximizing choice across five trade channels: direct shipment subject to tariffs, rerouting through a third country, de minimis shipment under the \$800 threshold, and ceasing exports altogether.

For each product m and potential origin country i , the firm compares the profit associated with each channel, accounting for tariffs, transport costs, and rerouting frictions. De minimis eligibility is captured through an indicator variable DM_m , and rerouting costs are allowed to vary with a composite measure of trade intensity $SIZE_{md}$. The firm selects the channel with the highest profit:

$$\text{Choice}_{idm} = \arg \max \left\{ \Pi_{idm}^{\text{direct}}, \max_r \Pi_{irmd}^{\text{reroute}}, \Pi_{idm}^{\text{de-minimis}}, 0 \right\}$$

This decision rule is applied to the actual observed trade war tariffs and to counterfactual scenarios (e.g., repeal of the de minimis exemption). Aggregate outcomes across products are used to evaluate changes in trade composition, tariff revenue, and channel substitution. A formal statement of the firm’s optimization problem is provided in Proposition 1 in Appendix C.

7 Counterfactual Simulations (In Progress)

The structural model provides a partial equilibrium framework to assess how exporters adjust across discrete shipment strategies in response to changes in trade policy and enforcement. Although general equilibrium feedbacks are abstracted from, the model captures the key margins relevant to tariff incidence, evasion behavior, and regulatory design.

One counterfactual eliminates the *de minimis* exemption by removing the low-value shipment option from the exporter’s set of choices. All goods must then enter through formal customs procedures and are subject to applicable tariffs. The simulation quantifies the volume of trade that would shift into taxed channels, the expected increase in tariff revenue, and the share of shipments that would no longer enter the market due to higher cost burdens. This scenario reflects current proposals to repeal or restrict the *de minimis* threshold and offers a direct estimate of the enforcement gap associated with this exemption.

A second scenario disables transshipment through third countries by assigning prohibitive costs to the rerouting option. With origin masking no longer viable, firms reallocate toward

direct shipment, fragmentation into small parcels, or exit. The simulated adjustment captures the extent to which observed trade diversion during the trade war relied on circumvention rather than genuine supplier substitution. These results are relevant for evaluating policies related to country-of-origin rules, anti-circumvention enforcement, and the integrity of preferential trade arrangements.

Further simulations raise the cost of using the *de minimis* channel to reflect enhanced enforcement. These adjustments represent regulatory changes such as mandatory advance data reporting, volume-based limits per recipient, or new screening requirements for express carriers. As administrative frictions increase, trade reallocates across other channels or falls due to exit. The response reveals the elasticity of circumvention behavior with respect to compliance costs and provides a benchmark for evaluating the expected effects of recent CBP rulemaking proposals.

A separate scenario removes all Section 301 trade war tariffs while holding all other frictions constant. This no-policy baseline isolates the impact of the tariff shock itself and provides a benchmark against which to assess the magnitude of circumvention behavior. The difference between observed and simulated flows under this setting reflects the extent to which firms altered their shipment strategies solely in response to new duties.

To evaluate the fiscal efficiency of tariff policy, the model is used to compute tariff revenue under a range of alternative duty rates. Simulated revenue initially increases with the tariff rate but declines once evasion margins become dominant. The result is a non-monotonic relationship between statutory tariff levels and realized collections, highlighting how leakage through *de minimis* shipments and transshipment reduces the effectiveness of high-duty regimes. The exercise underscores the importance of incorporating behavioral responses into optimal tariff design.

Finally, simulated welfare is compared across alternative enforcement environments, including full enforcement, observed evasion behavior, and the no-tariff baseline. While circumvention undermines revenue collection, it also reduces the burden passed through to consumers. The comparison reveals the trade-offs between administrative capacity, fiscal revenue, and consumer welfare. These findings contribute to broader debates on the distributional consequences of trade policy, particularly in settings where low-income consumers benefit disproportionately from tariff-exempt e-commerce channels.

The simulations as a whole offer a quantitative lens through which to evaluate enforcement intensity, evasion behavior, and the policy effectiveness of tariff instruments. By isolating key margins of firm-level adjustment, the framework provides a tractable method for estimating revenue losses, trade mismeasurement, and welfare implications arising from tariff circumvention.

8 Conclusion

The U.S.–China trade war prompted a wide array of adaptive behaviors as firms sought to mitigate tariff exposure. This analysis identifies two dominant circumvention channels: (i) shifting toward small-package *de minimis* shipments exempt from customs duties and (ii) rerouting non-*de minimis* goods through third countries not subject to Section 301 tariffs. These adjustments have broad implications for trade measurement, enforcement capacity, and the credibility of protectionist policy.

Using a new shipment-based measure of *de minimis* exposure constructed from proprietary CBP records, the paper shows that products previously traded through fragmented parcel channels experienced significantly larger declines in recorded U.S. imports following tariff increases. At the same time, these products exhibited weaker increases in measured duties paid and effective tariff rates, consistent with reduced exposure to formal customs collection. In contrast, products less compatible with fragmented parcel trade experienced stronger rerouting responses through intermediary economies and remained more exposed to conventional tariff incidence.

Together, the results point to a broader reorganization of trade flows during the trade war. Rather than adjusting exclusively through traditional supplier substitution, firms shifted across shipment and customs regimes in ways that reduced both measured imports and observed tariff collection. These patterns suggest that standard trade statistics may substantially understate true import activity when firms can fragment transactions into low-value parcels or reroute goods through third countries.

The findings also highlight an important limitation of modern trade enforcement. As cross-border e-commerce and direct-to-consumer logistics expand, an increasing share of international trade operates through fragmented shipment systems that receive simplified customs treatment and limited inspection. The rapid growth of *de minimis* trade therefore weakens both the fiscal incidence and the strategic effectiveness of tariff policy by allowing firms to reorganize transactions outside traditional enforcement channels.

The structural framework further illustrates how firms sort across export strategies in response to tariff shocks. Comparative statics show that higher tariffs increase the relative attractiveness of both rerouting and *de minimis* fragmentation, particularly for consumer-oriented products that are inexpensive, lightweight, and well suited to parcel delivery. These responses reduce realized tariff incidence and alter the observed geography of trade flows even when statutory tariff rates remain unchanged.

More broadly, the paper suggests that the effectiveness of trade policy depends not only on tariff levels, but also on the institutional boundaries of customs enforcement and the logistical organization of global commerce. As trade increasingly shifts toward high-frequency parcel

shipments and digitally coordinated fulfillment systems, traditional tariff instruments may become progressively less capable of targeting the intended flows of goods. Future research should examine the welfare, distributional, and enforcement implications of these changes and evaluate how customs modernization and regulatory reform can adapt to increasingly fragmented forms of international trade.

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Tables

Table 1: Top 20 HS-2 Categories Most Suitable for CBP De Minimis Entry (HS-6 basis)

Top 20	HS-2	Description	Number of HS-6 Products	CBP De Minimis HS-6	Share (%)
1	65	Headgear and parts thereof	8	7	87.5
2	61	Apparel and clothing accessories; knitted or crocheted	106	87	82.1
3	62	Apparel and clothing accessories; not knitted or crocheted	112	77	68.8
4	49	Printed books, newspapers, pictures and other products of the printing industry; manuscripts, typescripts and plans	19	13	68.4
5	42	Articles of leather; saddlery and harness; travel goods, handbags and similar containers; articles of animal gut	20	13	65.0
6	95	Toys, games and sports requisites; parts and accessories thereof	31	20	64.5
7	67	Prepared feathers and down; artificial flowers; articles of human hair	8	5	62.5
8	64	Footwear; gaiters and the like; parts of such articles	25	15	60.0
9	63	Other made-up textile articles; sets; worn clothing and worn textile articles; rags	51	29	56.9
10	94	Furniture; bedding, mattresses, mattress supports, cushions and similar stuffed furnishings; lamps and lighting fittings; illuminated signs; prefabricated buildings	39	21	53.8
11	96	Miscellaneous manufactured articles	48	22	45.8
12	50	Silk	9	4	44.4
13	97	Works of art, collectors' pieces and antiques	7	3	42.9
14	71	Natural or cultured pearls, precious or semi-precious stones, precious metals; imitation jewellery; coin	53	21	39.6
15	91	Clocks and watches and parts thereof	49	18	36.7
16	83	Miscellaneous articles of base metal	36	13	36.1
17	92	Musical instruments; parts and accessories of such articles	17	6	35.3
18	57	Carpets and other textile floor coverings	21	7	33.3
19	43	Furskins and artificial fur; manufactures thereof	12	4	33.3
20	66	Umbrellas, sun umbrellas, walking-sticks, seat sticks, whips, riding crops; and parts thereof	6	2	33.3
Total of the top 20			677	387	57.16
Total (all HS-2)			5206	737	14.16

Notes: Counts are numbers of HS-6 lines within each HS-2 category. “CBP De Minimis HS-6” counts HS-6 products flagged by the CBP de minimis indicator. Shares are CBP de minimis HS-6 divided by all HS-6 within the HS-2 category (%). Bottom rows report totals/averages for the top 20 and for all HS-2 categories.

Table 2: Characteristics of De Minimis and Non-De Minimis HS-6 Products

	De Minimis	Non-De Minimis	Difference
Air shipments	0.204	0.144	+0.060*** (5.80)
Vessel shipments	0.796	0.852	-0.056*** (-5.28)
Consumption good (BEC)	0.463	0.164	+0.299*** (18.91)
Intermediate good (BEC)	0.316	0.639	-0.323*** (-16.81)
Observations	732	3,963	

Notes: The table compares HS-6 products classified as de minimis eligible and non-de minimis eligible under the CBP de minimis indicator. “Air shipments” and “Vessel shipments” indicate whether the product was shipped by air or vessel using U.S. Census trade shipment data for 2016–2017. “Consumption good” and “Intermediate good” are based on the Broad Economic Categories (BEC) classification. The analysis uses a strict BEC mapping: consumption goods include only products classified as CONS, while intermediate goods include only products classified as INT. Mixed-use categories (e.g., CONS/INT, CAP/INT) are excluded from these indicators. The final column reports differences between de minimis and non-de minimis products. Robust *t*-statistics are reported in parentheses. *** denotes significance at the 1 percent level.

Table 3: Trade Flows Before vs. After July 2018, by CBP De Minimis Status

	De Minimis			Non-De Minimis		
	Obs.	Mean	Δ Mean	Obs.	Mean	Δ Mean
(i) U.S. Imports from China (log)						
Pre-July 2018	6,477	14.846		46,194	13.157	
Post-July 2018	5,024	14.782	-0.064	35,242	13.092	-0.065
(ii) Chinese Exports to Bystanders (log)						
Pre-July 2018	6,349	14.663		43,555	12.957	
Post-July 2018	4,987	14.595	-0.067	33,740	12.851	-0.106
(iii) U.S. Imports from Bystanders (log)						
Pre-July 2018	167,286	11.272		788,658	11.419	
Post-July 2018	133,293	11.295	+0.022	628,897	11.458	+0.039

Notes: The table reports monthly HS-6-level trade flows from January 2017 through August 2019. Products on List 4A are excluded. “Pre-July 2018” refers to months before July 2018; “Post-July 2018” includes July 2018 and all subsequent months in the sample. De minimis status is defined using the CBP de minimis indicator. U.S. imports are measured using U.S. Census trade data. “Bystanders” are non-China trading partners; the sample includes 235 countries. Δ Mean equals the post-period mean minus the pre-period mean, reported in log points.

Table 4: Summary Statistics for Tariff Exposure on Chinese Imports

Group	Obs.	Mean	Std. Dev.	Min	Max
Panel A. Full Sample					
<i>Pre-July 2018</i>	55,983	0.000	0.004	0.000	0.300
<i>Post-July 2018</i>	43,150	0.105	0.083	0.000	0.300
Non-De Minimis Products	87,330	0.046	0.076	0.000	0.300
De Minimis Products	11,803	0.043	0.074	0.000	0.250
Panel B. Excluding Steel, Aluminum, Solar, and Washer Tariffs					
<i>Pre-July 2018</i>	55,905	0.000	0.000	0.000	0.000
<i>Post-July 2018</i>	43,095	0.105	0.083	0.000	0.250
Non-De Minimis Products	87,197	0.046	0.076	0.000	0.250
De Minimis Products	11,803	0.043	0.074	0.000	0.250

Notes: The table reports descriptive statistics for tariff increases applied to Chinese imports at the monthly HS-6 level from January 2017 through August 2019. “Pre-July 2018” refers to months before July 2018, while “Post-July 2018” includes July 2018 and subsequent months. De minimis status is defined using the CBP de minimis indicator. Tariff increases are expressed as ad valorem rates (e.g., 0.25 = 25 percent). Panel A includes all tariff measures, including Section 301 tariffs, Section 232 steel and aluminum tariffs, and safeguard tariffs on solar panels and washing machines. Panel B excludes steel, aluminum, solar, and washer tariff measures, leaving only Section 301 tariff exposure.

Table 5: Direct Impact of Tariffs on U.S. Imports from China: OLS Estimates

	Value (1)	Quantity (2)	Unit Value (3)	Price (Incl. Duty) (4)
<i>Baseline</i>				
$\log(1 + \tau)$	-1.839*** (0.098)	-2.028*** (0.140)	0.213*** (0.080)	0.975*** (0.081)
<i>+ De minimis interaction</i>				
$\log(1 + \tau)$	-1.786*** (0.095)	-2.060*** (0.141)	0.260*** (0.083)	1.018*** (0.083)
De minimis $\times \log(1 + \tau)$	-0.440* (0.244)	0.258 (0.294)	-0.378*** (0.141)	-0.347** (0.143)
Observations	255,729	227,104	227,104	227,104
<i>Fixed effects: Month and HS10</i>				
<i>Standard errors: Clustered by HS8</i>				

Notes: Table reports OLS estimates for monthly U.S. imports from China at the HS10-product level. The dependent variables are log import value, log import quantity, log unit value, and log tariff-inclusive price. $\log(1 + \tau)$ denotes the log tariff change, where τ is the ad valorem tariff rate increase. “De minimis” is defined using the CBP de minimis indicator. All specifications include month and HS10 fixed effects. Standard errors clustered at the HS8 level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Direct Impact of Tariffs on U.S. Imports from China: Robustness to Alternative De Minimis Proxies

	CBEC List Indicator (1)	Under \$800 Unit Value (2)	Consumption Strict (3)	Consumption Broad (4)	Air Common Product (5)
Panel A. Import Value					
$\log(1 + \tau_{it})$	-4.785*** (0.960)	-2.599*** (0.997)	-5.236*** (0.986)	-5.101*** (0.969)	-3.614*** (1.236)
Proxy $\times \log(1 + \tau_{it})$	-4.772** (2.014)	-3.108*** (0.893)	1.030 (1.811)	-3.988 (2.774)	-11.300** (5.737)
Observations	255,729	254,410	255,654	255,654	255,654
HS8 clusters	7,039	6,968	7,014	7,014	7,014
Panel B. Import Quantity					
$\log(1 + \tau_{it})$	-3.969 (2.996)	-1.702 (1.117)	-4.259 (2.884)	-4.300 (2.913)	-2.846* (1.586)
Proxy $\times \log(1 + \tau_{it})$	-2.979 (3.097)	-3.164 (2.405)	0.370 (2.263)	1.362 (1.923)	-12.000 (17.300)
Observations	227,104	226,812	227,029	227,029	227,029
HS8 clusters	6,636	6,587	6,611	6,611	6,611
Panel C. Unit Value					
$\log(1 + \tau_{it})$	-2.370 (3.280)	-27.200* (14.200)	-2.000 (3.190)	-2.130 (3.200)	-0.733 (2.670)
Proxy $\times \log(1 + \tau_{it})$	3.700 (2.450)	31.100** (14.800)	4.230* (2.320)	4.060* (2.330)	-10.800 (13.400)
Observations	227,104	226,812	227,029	227,029	227,029
HS8 clusters	6,636	6,587	6,611	6,611	6,611
Panel D. Tariff-Inclusive Price					
$\log(1 + \tau_{it})$	4.340 (3.460)	-27.200* (14.200)	4.330 (3.350)	4.310 (3.370)	-0.733 (2.670)
Proxy $\times \log(1 + \tau_{it})$	-0.113 (2.560)	31.100** (14.800)	1.240 (2.370)	0.702 (2.430)	-10.800 (13.400)
Observations	227,104	226,812	227,029	227,029	227,029
HS8 clusters	6,636	6,587	6,611	6,611	6,611

Notes: Table reports OLS estimates for monthly U.S. imports from China at the HS10-product level. Outcomes are import value, quantity, unit value, and tariff-inclusive price. Columns (1)–(5) sequentially replace the baseline CBP de minimis indicator with alternative proxies for de minimis suitability. CBEC indicates products included on China’s cross-border e-commerce positive list. Under \$800 identifies products with 2017 unit values below the U.S. de minimis threshold. Consumption strict and broad use BEC classifications. Air common identifies products frequently shipped by air. All specifications include HS10 and month fixed effects. Standard errors clustered at the HS8 level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Tariff Incidence by De Minimis Eligibility: OLS Estimates

	Log Duties Paid		Effective Tariff Rate	
	Baseline	Interaction	Baseline	Interaction
$\log(1 + \tau_{it})$	7.423*** (0.251)	7.579*** (0.281)	9.312*** (0.229)	9.681*** (0.258)
De minimis $\times \log(1 + \tau_{it})$		-0.903** (0.411)		-2.122*** (0.395)
Observations	101,071	101,071	101,071	101,071
HS6 clusters	3,576	3,576	3,576	3,576

Fixed effects: HS6 and month

Standard errors: Clustered by HS6

Notes: The table reports OLS estimates for monthly U.S. imports from China at the HS6-product level. Columns (1)–(2) use log duties paid as the dependent variable. Columns (3)–(4) use the log effective tariff rate, defined as $\log(\text{duties paid}/\text{import value})$. Tariff exposure is measured using $\log(1 + \tau_{it})$, where τ_{it} is the applied MFN plus Section 301 tariff rate. “De minimis” is defined using the CBP de minimis indicator. All regressions include HS6 and month fixed effects. Standard errors clustered by HS6 are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Rerouting Through Third Countries: Quarterly OLS Estimates

	Iyoha et al. (2024) Measure		Relative Trade Flow Measure	
	(1)	(2)	(3)	(4)
Tariff measure	τ_{it}	τ_{it}	$\log(1 + \tau_{it})$	$\log(1 + \tau_{it})$
	0.032***	0.038***	0.233***	0.274***
	(0.008)	(0.008)	(0.076)	(0.077)
De minimis $\times$ tariff		-0.036**		-0.259**
		(0.015)		(0.127)
Observations	743,106	743,106	552,296	552,296
HS6 clusters	3,836	3,836	3,309	3,309
<i>Fixed effects:</i> HS6 $\times$ partner and partner $\times$ quarter				
<i>Standard errors:</i> Clustered by HS6				

Notes: The table reports quarterly OLS estimates of rerouting through third countries. Columns (1)–(2) use the rerouting-share measure from Iyoha et al. (2024):

$$\text{Reroute}_{ipt} = \frac{\min(X_{it}^{CHN \rightarrow p}, M_{it}^{US \leftarrow p})}{M_{it}^{US \leftarrow p}}$$

where $X_{it}^{CHN \rightarrow p}$ denotes Chinese exports to bystander country p , and $M_{it}^{US \leftarrow p}$ denotes U.S. imports from bystander p for product i in quarter t . Columns (1)–(2) use the applied MFN plus Section 301 tariff level, τ_{it} .

Columns (3)–(4) use a relative trade-flow measure defined as:

$$\log(X_{it}^{CHN \rightarrow p}) - \log(M_{it}^{US \leftarrow p})$$

and use $\log(1 + \tau_{it})$ as the tariff measure.

“De minimis” is defined using the CBP de minimis indicator. All regressions include HS6 $\times$ partner and partner $\times$ quarter fixed effects. Standard errors clustered by HS6 are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Rerouting Through Third Countries: Robustness to Alternative De Minimis Proxies

	CBEC List Indicator (1)	Under \$800 Unit Value (2)	Consumption Strict (3)	Consumption Broad (4)	Air Common Product (5)
MFN + Section 301 tariff, τ_{it}	0.036*** (0.008)	0.053*** (0.018)	0.031*** (0.008)	0.032*** (0.008)	0.032*** (0.008)
Proxy $\times \tau_{it}$	-0.031*** (0.011)	-0.027 (0.019)	-0.058*** (0.020)	-0.011 (0.020)	0.002 (0.017)
Observations	743,106	724,936	735,834	735,834	735,834
HS6 clusters	3,836	3,576	3,625	3,625	3,625

Fixed effects: HS6 $\times$ partner and partner $\times$ quarter

Standard errors: Clustered by HS6

Notes: The dependent variable is the quarterly rerouting-share measure from Iyoha et al. (2024), defined as $\min(X_{it}^{CHN \rightarrow p}, M_{it}^{US \leftarrow p})/M_{it}^{US \leftarrow p}$, where $X_{it}^{CHN \rightarrow p}$ denotes Chinese exports to bystander country p , and $M_{it}^{US \leftarrow p}$ denotes U.S. imports from bystander p . The sample excludes China as a partner country. Column (1) uses China's cross-border e-commerce (CBEC) positive list indicator. Column (2) defines products as de minimis-suitable if their 2017 U.S. imports from China had unit values below \$800 using UN Comtrade data. Columns (3)–(4) use BEC consumption-good classifications: the strict measure includes only products classified as CONS, while the broad measure also includes mixed-use categories containing consumption use. Column (5) indicates products commonly shipped by air using U.S. Census trade shipment data for 2016–2017. All regressions include HS6 $\times$ partner, quarter, and partner $\times$ quarter fixed effects. Standard errors clustered by HS6 are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Rerouting Through Third Countries: Robustness to Product-Trait Quintiles

	Low Air Share Quintiles (1)	Trade Cost Quintiles (2)	Unit Value Quintiles (3)
MFN + Section 301 tariff, τ_{it}	0.044*** (0.013)	0.040*** (0.012)	0.047*** (0.014)
Q2 $\times \tau_{it}$	-0.013 (0.015)	0.007 (0.014)	-0.026 (0.017)
Q3 $\times \tau_{it}$	-0.003 (0.015)	-0.013 (0.016)	-0.039** (0.016)
Q4 $\times \tau_{it}$	-0.023 (0.019)	-0.022 (0.018)	-0.017 (0.018)
Q5 $\times \tau_{it}$	-0.043** (0.022)	0.000 (0.018)	0.003 (0.016)
Observations	642,843	642,843	642,843
HS6 clusters	3,338	3,338	3,338
<i>Fixed effects:</i> HS6 $\times$ partner, quarter, and partner $\times$ quarter			
<i>Standard errors:</i> Clustered by HS6			

Notes: The dependent variable is the quarterly rerouting-share measure from Iyoha et al. (2024), defined as $\min(X_{it}^{CHN \rightarrow p}, M_{it}^{US \leftarrow p})/M_{it}^{US \leftarrow p}$. Each column reports one regression interacting MFN plus Section 301 tariff exposure with HS6-level quintiles constructed from pre-trade-war U.S.–China data. Q1 is the omitted category. Low Air Share is defined as 6 minus the air-shipment-share quintile, so higher quintiles indicate products less commonly shipped by air. Trade Cost is based on `charges_imp`. Unit Value is value divided by quantity. Quintiles are defined using products with nonmissing unit values. All regressions include HS6 $\times$ partner, quarter, and partner $\times$ quarter fixed effects. Standard errors clustered by HS6 are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figures

US-China Trade War Timeline

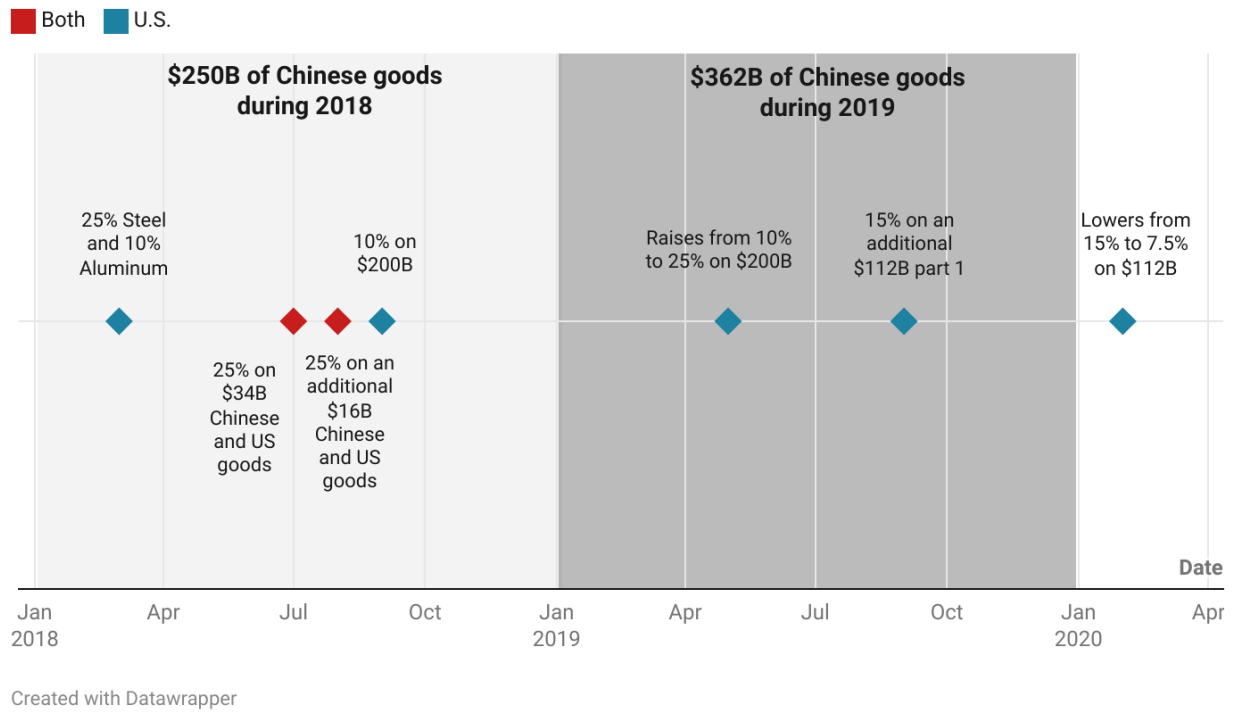
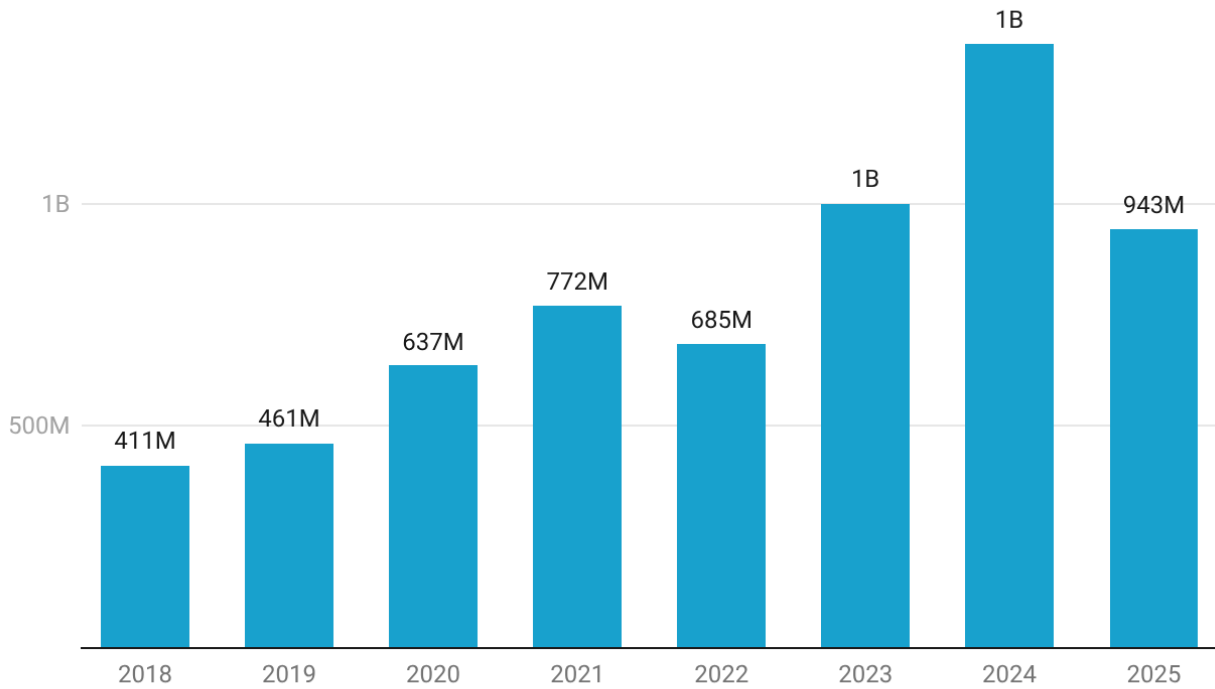


Figure 1: US-China Trade War Timeline

De Minimis Total Volumes (2018–2025)

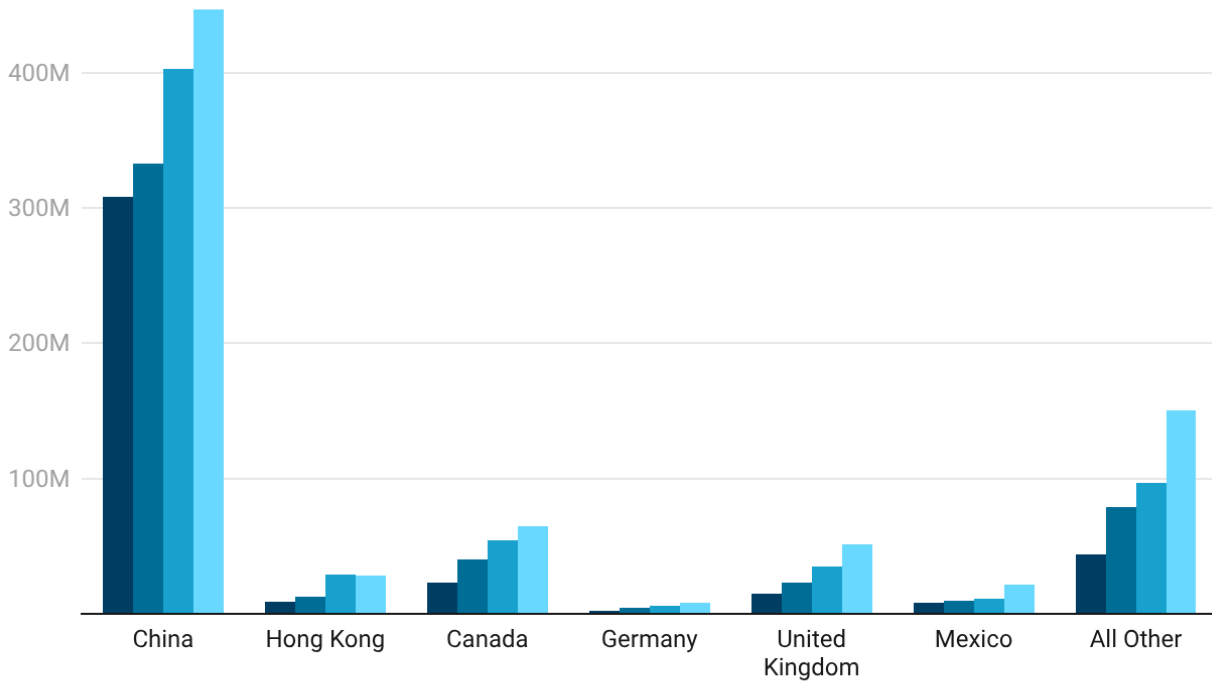


Created with Datawrapper

Figure 2: De Minimis Shipment Volumes

De Minimis Volume by Country of Shipment (2018–2021)

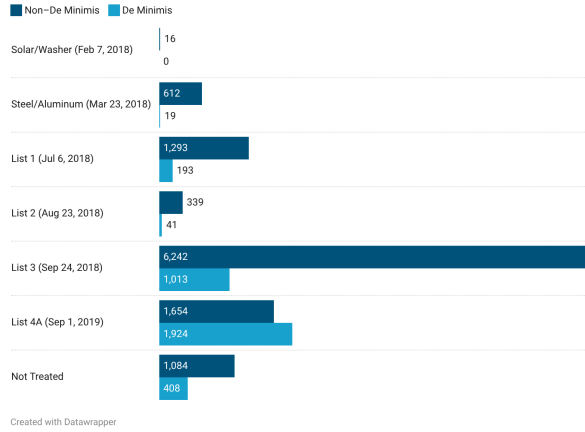
2018 2019 2020 2021



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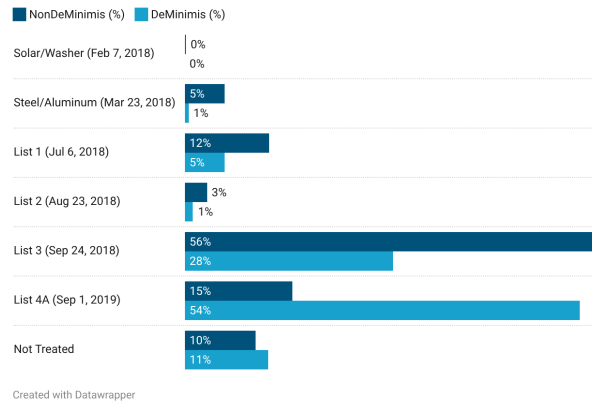
Figure 3: De Minimis Shipment Volumes, by Country of Origin

Tariff Exposure Timing by De Minimis Status (China, HS10 Level)



(a) Counts

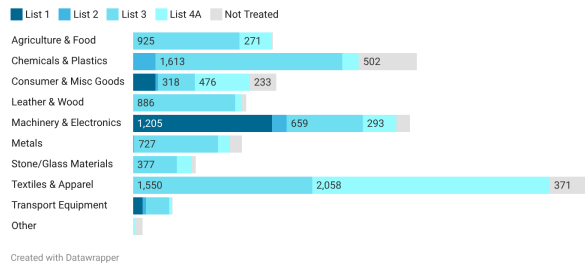
Distribution of Products by Tariff Wave and De Minimis Status



(b) Shares

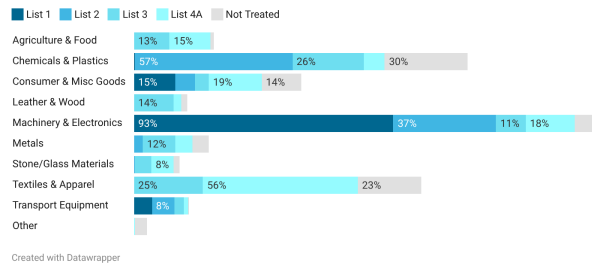
Figure 4: Tariff exposure timing by de minimis status (China, HS10 level). Left panel shows counts of products by tariff wave. Right panel shows the share of products within each group (de minimis and non-de minimis) across tariff waves.

Distribution of Products Across Tariff Waves by Category, Counts



(a) Counts

Distribution of Products Across Tariff Waves by Category, Shares



(b) Shares

Figure 5: Distribution of products across tariff waves by category. Panel (a) reports the number of products in each category across tariff waves, while Panel (b) reports the corresponding shares within each wave. Later tariff waves, particularly List 4A, are disproportionately composed of textiles and consumer goods, whereas earlier waves are dominated by machinery and industrial inputs.

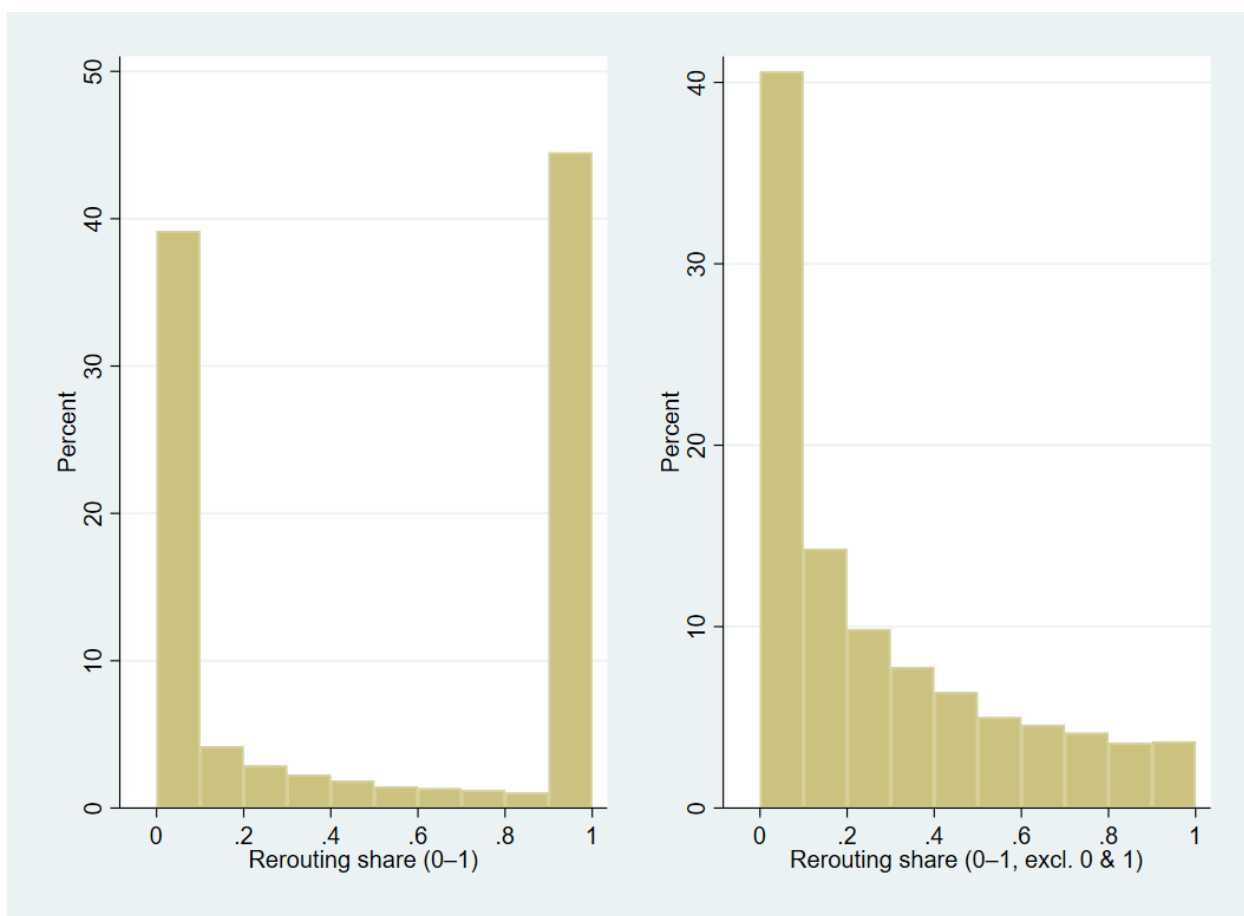
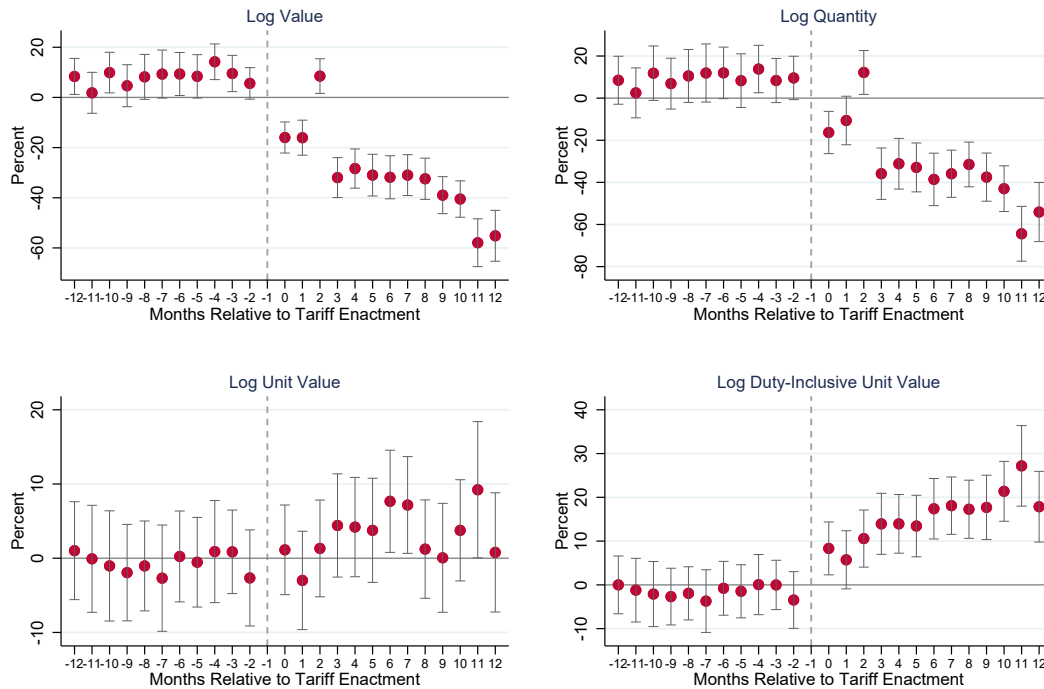


Figure 6: Distribution of Product-Level Rerouting Shares

Sun-Abraham Event Study: Non-De Minimis



Sun-Abraham Event Study: De Minimis

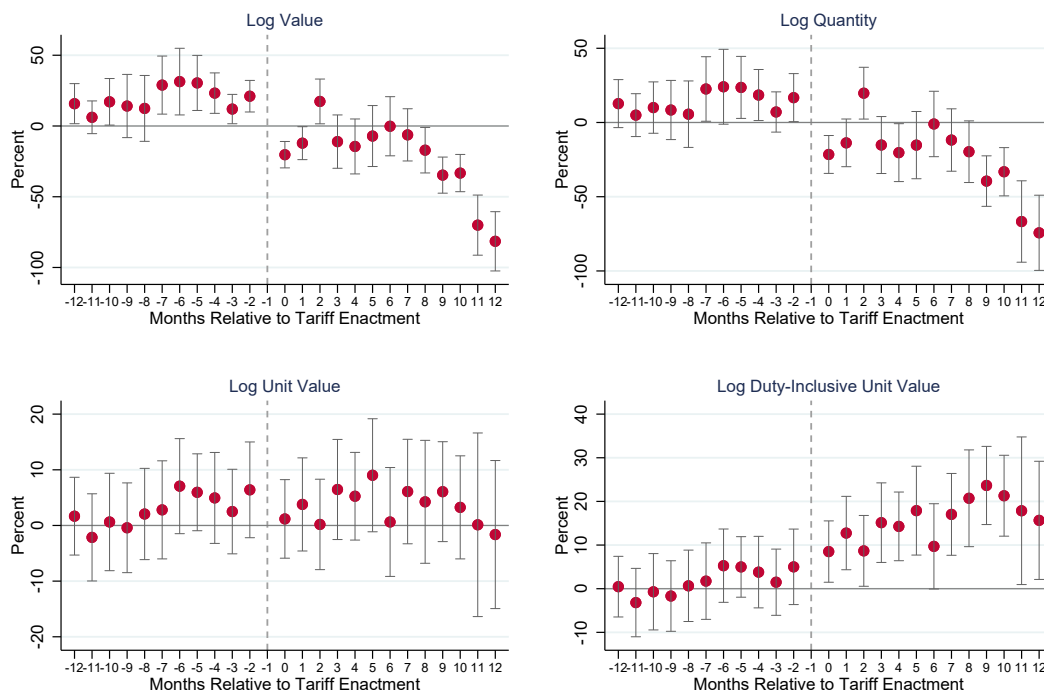


Figure 7: Event Study: Timing and Pre-Trends

Appendix

A Construction of the De Minimis Product Measure

A.1 CBP Shipment Coverage

The CBP dataset contains transaction-level records for shipments arriving during the first week of December in each year from 2017 through 2021. The sample includes shipments with declared values up to \$1,500. Using CBP shipment-level records matched to 6-digit HTS codes through cargo descriptions, 736 HTS products are identified in 2017 that (i) had declared values between \$0 and \$800 (the de minimis threshold) and (ii) shipped at least 25 packages during the sample week. In total, the 2017 CBP data contain 251,055 China-origin shipments with declared values between \$0 and \$800.

A.2 Chinese Shipment Growth Over Time

Table 11 reports the number of non-zero China-origin shipments observed in the CBP sample by year.

Table 11: China-Origin Shipments Over Time

Year	All Shipments ($\leq 1,500$)	De Minimis (≤ 800)
2017	258,018	254,064
2018	391,043	388,887
2019	476,348	474,449
2020	852,595	851,996
2021	846,627	845,777
Total	2,824,631	2,815,173

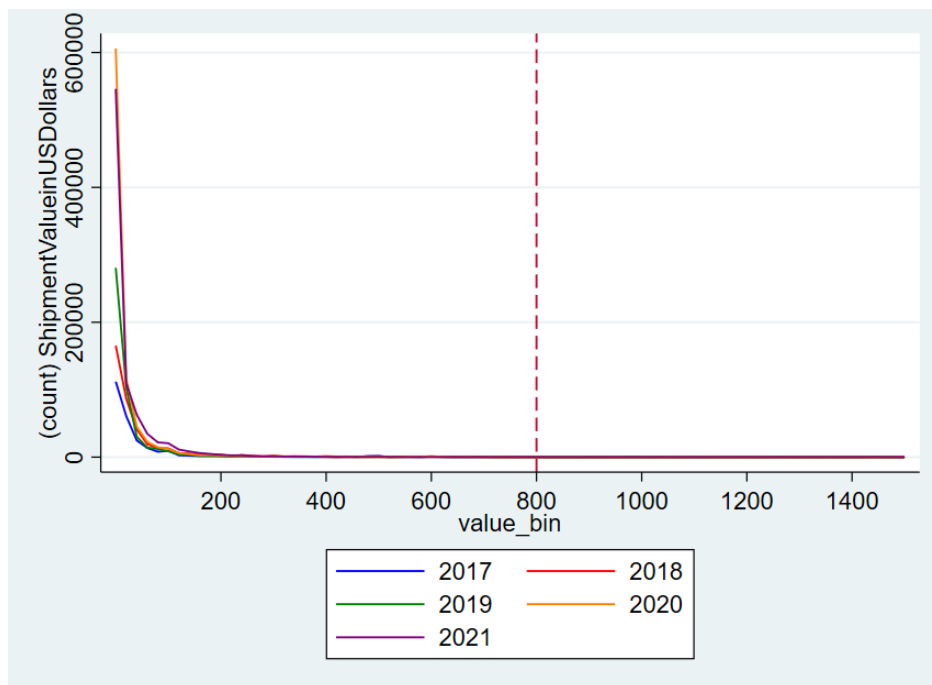
Notes: Shipments restricted to positive declared values and Country of Origin = China. The sample includes shipments arriving during the first week of December each year (2017–2021).

China-origin shipments increase steadily from 2017 to 2019 and then nearly double between 2019 and 2020. The sharp expansion coincides with the post-trade war adjustment period and rapid growth in cross-border e-commerce channels.

A.3 Shipment Value Distribution

Shipment values for China-origin shipments exhibit substantial right-skewness. Including zero values, the median shipment value in 2017 is \$22, while the mean is \$75.27, indicating a heavy right tail. The 99th percentile exceeds \$1,050 and kurtosis is approximately 29, confirming extreme outliers.

Figure 8 plots the distribution of shipment values by year. The red dashed line denotes the \$800 de minimis threshold.



(a) Distribution of Shipment Values by Year (China-Origin)

Figure 8: Shipment Value Distribution, 2017–2021. The dashed line denotes the \$800 de minimis threshold.

Figure 1 shows that the distribution is highly concentrated at low declared values in all years. It also shows that the mass of shipments below \$800 grows substantially after 2019.

B CBP 2017 Sample Overview

The transaction-level dataset contains a total of 1,048,575 shipments recorded between December 4 and December 10, 2017. Table 12 presents the daily distribution of transactions.

Table 12: Daily Distribution of Transactions: Overall and China Shipments

Date	All Shipments		China (CN) Shipments	
	Frequency	Percent (%)	Frequency	Percent (%)
04 Dec 2017	134,797	12.86	33,993	13.08
05 Dec 2017	180,180	17.18	47,158	18.14
06 Dec 2017	167,973	16.02	33,207	12.78
07 Dec 2017	195,940	18.69	41,577	16.00
08 Dec 2017	191,487	18.26	44,963	17.30
09 Dec 2017	99,575	9.50	29,039	11.17
10 Dec 2017	78,623	7.50	30,000	11.54
Total	1,048,575	100.00	259,937	100.00

Transaction volume peaks on December 7–8 and declines over the weekend (December 9–10), suggesting a strong weekday concentration in shipment activity. The daily distribution for China-origin shipments closely mirrors the overall pattern, though China shipments exhibit a relatively higher share over the weekend (December 9–10), indicating a slightly less pronounced weekday concentration compared to the full sample.

B.1 Country of Origin Distribution

The dataset includes shipments from a wide range of countries. However, the distribution is highly concentrated.

China (CN) accounts for 259,937 shipments (24.79%), followed by the United Kingdom (15.38%) and Hong Kong (11.18%). A substantial proportion of observations (31.69%) have missing country-of-origin information (coded as “None”).

When restricting the sample to positive-value shipments at or below the \$800 de minimis threshold ($0 < \text{ShipmentValue} \leq 800$), the share of missing country-of-origin observations declines sharply from 31.69% to 1.80%. Within this restricted sample, China accounts for 36.3% of shipments (excluding “None”), followed by the United Kingdom (22.9%) and Hong Kong (16.7%). Canada accounts for 2.8%, while Germany and Mexico account for approximately 1–1.5% each.

While China remains the largest origin country in both the full and restricted samples, its share in this 2017 dataset (36.3% in the de minimis-restricted sample) is lower than the approximately 64% share reported in U.S. Customs and Border Protection (CBP) fiscal year 2018–2021 de minimis statistics. Conversely, the United Kingdom and Hong Kong are substantially overrepresented relative to CBP aggregates.

These differences likely reflect a combination of (i) the December 2017 time window of

the sample, (ii) carrier- or channel-specific coverage rather than universe-level entry data, and (iii) structural changes in de minimis trade flows after 2018, during which China’s share increased substantially. Accordingly, while the sample replicates the qualitative dominance of China as the primary origin, it exhibits greater geographic diversification than later CBP administrative totals.

B.2 Embedding Confidence

Embedding confidence scores range approximately from 0.15 to 0.85. Summary statistics for the full sample are presented in Table 13.

Table 13: Embedding Confidence – Full Sample

Statistic	Value
Observations	1,048,218
Mean	0.482
Median	0.483
Std. Dev.	0.084
25th Percentile	0.420
75th Percentile	0.537

The distribution is approximately symmetric with mild dispersion.

For shipments originating from China (CN), the mean confidence is slightly higher (0.489) and exhibits slightly lower variance (0.0063 versus 0.0071 in the full sample). The distribution shows mild negative skewness (-0.185), suggesting slightly more mass in the upper tail relative to the lower tail.

B.3 Product Composition

Table 14 reports the top 25 HTS codes with the highest number of packages in the 2017 CBP de minimis sample.

HTS	Description (Abbreviated)	Packages
611020	Cotton sweaters/pullovers (knit)	18,781
620444	Women’s dresses, artificial fibers	11,569
610910	Cotton T-shirts (knit)	11,316
620640	Women’s blouses, man-made fibers	6,337
847130	Portable computers (≤10kg)	5,763
620442	Women’s dresses, cotton	5,231
620431	Women’s wool jackets/blazers	4,657
620630	Women’s cotton blouses	4,408
610443	Women’s synthetic dresses (knit)	4,364
621050	Women’s garments (special fabrics)	4,255
610442	Women’s cotton dresses (knit)	4,020
950300	Toys and models	3,988
620719	Men’s underpants (non-cotton)	3,826
610610	Women’s cotton blouses (knit)	3,727
670411	Synthetic wigs	3,501
711790	Imitation jewelry	3,387
611011	Wool sweaters (knit)	3,173
851712	Cellular telephones	3,138
640411	Textile sports footwear	3,072
611120	Babies’ cotton garments (knit)	2,960
640320	Leather footwear (strap)	2,925
610729	Men’s knit nightwear (other)	2,331
621120	Ski suits	2,313
610120	Men’s cotton coats (knit)	2,292
711320	Jewelry, base metal clad	1,988

Table 14: Top 25 HTS codes by number of packages in CBP 2017 (de minimis shipments).

The distribution is heavily concentrated in apparel, footwear, jewelry, electronics, and toys, indicating that low-value cross-border e-commerce shipments are dominated by consumer goods categories.

B.4 Overlap with Trade War Tariff Targeting

During the U.S.–China trade war, the majority of HTS codes were eventually targeted by Section 301 tariffs. Table 15 compares the 2017 CBP de minimis HTS codes to whether those codes were ever targeted.

Trade War Targeted	CBP 2017 (De Minimis)		
	No	Yes	Total
Not Targeted	195	52	247
Targeted	4,273	684	4,957
Total	4,468	736	5,204

Table 15: Comparison of 2017 CBP HTS codes with trade war tariff targeting.

Of the 736 HTS codes observed in the 2017 de minimis sample, 684 (93%) were eventually subject to tariffs. Thus, the de minimis shipment margin largely overlaps with products that became tariff-targeted.

B.5 Comparison of CBP and CBEC De Minimis Classifications

This section compares the de minimis classification based on U.S. Customs and Border Protection (CBP) shipment data with an alternative classification derived from the cross-border e-commerce (CBEC) product list. While both measures aim to capture goods likely to be traded through small-parcel channels, they differ substantially in construction and economic interpretation.

Table 16 reports the overlap between the two classifications across the full set of HS6 products. Of the 737 products classified as de minimis using CBP data, 378 (51.3 percent) are also classified as CBEC products, while 359 (48.7 percent) are not. Conversely, a large share of CBEC products are not identified as de minimis in the CBP data: 729 out of 1,107 products (65.9 percent) fall into this category.

Table 16: Overlap Between CBP and CBEC Classifications

Sample	CBP	CBEC Classification		
		Non-Ecommerce	Ecommerce	Total
All HS6	Non-De Minimis	3,740	729	4,469
	De Minimis	359	378	737
	Total	4,099	1,107	5,206
China-U.S. Trade	Non-De Minimis	3,500	712	4,212
	De Minimis	358	378	736
	Total	3,858	1,090	4,948

Table 16 also restricts the sample to products that are traded between China and the

United States. The patterns are similar: among CBP *de minimis* products, 378 out of 736 (51.4 percent) are also classified as CBEC, while 358 (48.6 percent) are not. At the same time, a substantial number of CBEC products (712 out of 1,090, or 65.3 percent) are not identified as *de minimis* in the CBP data.

These results indicate only partial overlap between the two classifications. Roughly half of the CBP-identified *de minimis* products are captured by the CBEC list, but the CBEC classification includes a large number of products that are not observed in low-value shipments in the CBP data.

To further assess the differences, Figure 16 examines the distribution of products that are classified as CBEC but not *de minimis* in the CBP data. These products are broadly distributed across HS chapters, with no clear concentration in categories typically associated with small, lightweight consumer goods. Instead, they span a wide range of sectors, including machinery (HS84), electronics (HS85), chemicals, and various manufactured goods.

Moreover, a non-trivial share of these products belong to food and agricultural categories (HS chapters below 28), including items such as dairy products, prepared foods, beverages, and meat products. These goods are often subject to sanitary regulations, import restrictions, and additional taxation in the United States, making them unlikely candidates for systematic shipment through *de minimis* channels.

In contrast, products identified as *de minimis* in the CBP data exhibit a much stronger concentration in categories consistent with small-package trade, particularly apparel (HS61–62), electronics (HS85), and other consumer goods. This pattern aligns with the shipment-based definition and reinforces the interpretation of the CBP measure as capturing actual trade behavior.

Taken together, these findings suggest that the CBEC classification is a noisy proxy for *de minimis* exposure. While it captures some overlap with CBP-identified products, it also includes a substantial number of goods that are unlikely to be traded through low-value shipment channels. As a result, the CBP-based classification provides a more precise measure of *de minimis* trade exposure for the purposes of the empirical analysis.

C Proof

Proposition 1: Conditions for Rerouting Non-De Minimis Goods

Consider a firm producing product m in China that is not eligible for *de minimis* treatment or for which shipment size exceeds the feasible parcel threshold, $\text{SIZE}_{md} > \underline{S}$. The firm chooses between shipping directly to the United States through standard customs channels (Direct Non-*De Minimis*, Channel ii) and rerouting through a bystander country b (Rerout-

ing through Bystander, Channel iii). For a given quantity Q , the firm selects the channel that maximizes profits. Rerouting is preferred whenever

$$\pi_m^{\text{reroute}}(Q) > \pi_m^{\text{dir}}(Q).$$

Substituting the profit functions from the main text yields

$$\begin{aligned} p_m^{\text{US}} Q - c_m Q - (p_m^{\text{US}} \cdot \tau_m^{\text{CN},b}) Q - (p_m^{\text{US}} \cdot \tau_m^{b,\text{US}}) Q - t_m^{\text{reroute}}(s_m, \phi_m) Q - f_m^{\text{reroute}} \\ > p_m^{\text{US}} Q - c_m Q - (p_m^{\text{US}} \cdot \tau_m^{\text{US}}) Q - t_m^{\text{dir}}(s_m, \phi_m) Q. \end{aligned}$$

Dividing both sides by Q and rearranging terms implies that rerouting is optimal whenever

$$p_m^{\text{US}} \cdot \tau_m^{\text{US}} + t_m^{\text{dir}}(s_m, \phi_m) > p_m^{\text{US}} \cdot (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}}) + t_m^{\text{reroute}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q}.$$

Equivalently, rerouting is preferred when

$$p_m^{\text{US}} [\tau_m^{\text{US}} - (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}})] > t_m^{\text{reroute}}(s_m, \phi_m) - t_m^{\text{dir}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q}. \quad (\text{P1})$$

Proof of Proposition 1:

The firm's decision to reroute non-*de minimis* goods is determined by a comparison of generalized costs across export channels. Let the generalized per-unit cost of direct shipment be

$$G_m^{\text{dir}} = p_m^{\text{US}} \tau_m^{\text{US}} + t_m^{\text{dir}}(s_m, \phi_m),$$

and let the generalized per-unit cost of rerouting be

$$G_m^{\text{re}} = p_m^{\text{US}} (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}}) + t_m^{\text{reroute}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q}.$$

Because production costs c_m and revenues $p_m^{\text{US}} Q$ are common across the two channels, the firm's optimization problem reduces to minimizing generalized trade costs. Rerouting is therefore optimal whenever

$$G_m^{\text{re}} < G_m^{\text{dir}}.$$

Substituting the generalized cost expressions yields

$$p_m^{\text{US}} (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}}) + t_m^{\text{reroute}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q} < p_m^{\text{US}} \tau_m^{\text{US}} + t_m^{\text{dir}}(s_m, \phi_m).$$

Rearranging terms produces condition (P1):

$$p_m^{\text{US}} [\tau_m^{\text{US}} - (\tau_m^{\text{CN},b} + \tau_m^{b,\text{US}})] > t_m^{\text{reroute}}(s_m, \phi_m) - t_m^{\text{dir}}(s_m, \phi_m) + \frac{f_m^{\text{reroute}}}{Q}.$$

This inequality demonstrates that rerouting becomes optimal when the tariff savings from transshipment through a bystander country exceed the additional transportation and fixed costs associated with rerouting. The condition depends on the tariff differential across routes, shipment size s_m , transport suitability ϕ_m , and operating scale Q , which determines the per-unit burden of the fixed rerouting cost.

The condition also highlights the distinction between rerouting and *de minimis* shipment emphasized in the main text. Rerouting is most attractive for products shipped at sufficiently large scale and well suited to bulk shipment networks. By contrast, products with high value from rapid shipment and inventory flexibility may instead prefer fragmented *de minimis* shipment channels, consistent with the fast-transport mechanism emphasized by Hummels and Schaur (2010).